

Revisiting Fundamentals of the European Gas Market: The Role of Supply Substitution

Abstract

We study the European natural gas market through a Bayesian Structural Vector Autoregression that distinguishes pipeline gas from liquefied natural gas (LNG), with contemporaneous elasticities identified through sign and elasticity restrictions. Pipeline gas is highly price-inelastic and propagates supply disruptions as transient spikes, whereas LNG acts as the flexible balancing margin with more persistent price effects. Three counterfactual exercises quantify the contributions of supply substitution and demand adjustment during the 2022 energy crisis. Preventing European LNG imports from expanding during the crisis raises the real price by roughly 125% relative to its observed path. At the opposite extreme, allowing LNG imports to fully absorb the contemporaneous pipeline-and-domestic shortfall lowers prices by about 82%. Eliminating the autonomous demand-reduction signal identified by the model over the formal EU policy window raises prices by approximately 35%. The realised path lies closer to the LNG-frozen upper bound than to the LNG-unconstrained lower bound, consistent with Europe’s limited regasification capacity binding during the crisis. A forward-looking scenario fan applied to the 2026 Strait of Hormuz crisis shows that the price impact depends sharply on whether the disruption resolves promptly or persists: median twelve-month price gaps range from roughly -5% under prompt recovery to $+34\%$ under a chronic disruption and $+127\%$ if a second wave materialises. The asymmetry underscores geographic diversification of LNG supply as a complementary energy-security instrument alongside receiving-side infrastructure.

1 Introduction

The 2022 European energy crisis brought the role of supply composition in natural gas markets into unusually sharp relief. Within a few months, deliveries through the Russian pipeline network collapsed from more than 40% of total European net imports to a residual share, and liquefied natural gas (LNG) imports rose by over 60% to partly absorb the resulting shortfall. The composition of European gas supply shifted more between February 2022 and December 2023 than over the preceding decade. The episode raised three questions for which the existing empirical literature is poorly equipped: how much of the price spike was attributable to the supply disruption itself, and how much to the inability of the substitute margin to scale up fast enough; what would prices have been under alternative LNG paths; and how much of the realised stabilisation came from supply substitution rather than from autonomous demand reduction.

The macroeconomic stakes of these questions are first-order. Natural gas is a substantial component of European headline inflation and was the dominant proximate driver of the surge in euro-area HICP from 5.1% in January 2022 to a peak of 10.6% in October 2022. The European Central Bank’s response, the fastest tightening cycle in its history, raised the deposit-facility rate from -0.5% to 4.0% between July 2022 and September 2023, with energy-price persistence cited repeatedly as a central concern in policy communications

(Adolfson et al., 2024). Whether a given gas-market disturbance generates transient or persistent inflationary pressure depends in turn on the structural composition of the supply disruption: pipeline disruptions transmit, on the evidence we report below, as relatively short-lived price spikes that the SVAR identifies as largely dissipating within five months, whereas LNG-supply shocks shift the medium-run market-clearing condition and propagate more persistently. Distinguishing the two transmission channels is therefore not only an academic refinement but a precondition for the calibration of monetary-policy responses to gas-market disruptions, of which the ongoing 2026 Strait of Hormuz episode is the most recent illustration.

Natural gas accounts for roughly 25% of total energy consumption in the European Union and is predominantly imported. Two physically and contractually distinct channels deliver these imports. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay and destination clauses, and is constrained by fixed transport infrastructure with limited spare capacity. LNG, by contrast, is increasingly traded on shorter-duration spot or hub-indexed contracts, with cargoes that can be redirected across regions in response to price differentials (Sharples, 2023; Ritz, 2019; Molnar, 2022). The two channels differ markedly in short-run elasticity, adjustment cost and contractual rigidity. Until 2018, pipeline gas accounted for over 85% of European net imports; by 2024 its share had fallen below 60%. This structural shift in supply composition, together with the abrupt and asymmetric nature of the 2022 disruption, makes the explicit identification of supply by source not a refinement of existing models but a precondition for sensible inference about gas-price formation.

Most empirical work on natural gas adapts the Structural Vector Autoregression (SVAR) framework developed for the crude-oil market by Kilian (2009); Kilian and Murphy (2014); Baumeister and Hamilton (2019), in which supply is treated as a single aggregate. In oil this is a defensible approximation; in gas, where the two supply channels respond differently to the same shocks and where the share of each has been shifting structurally, it is not. Aggregation conceals the compositional adjustment that drove the 2022 dynamics and obscures the heterogeneous transmission of supply shocks across horizons. Recent contributions emphasise that ignoring this margin systematically understates the role of supply-side forces (Moll et al., 2023; Hamilton, 2023), and that the 2022 crisis was as much a substitution episode as a level-shock episode (Güntner et al., 2024). The contribution of this paper is to model the two supply channels explicitly within a Bayesian SVAR framework that delivers economically meaningful identification of their contemporaneous elasticities.

The identification strategy follows the Bayesian sign- and elasticity-restriction approach of Baumeister and Hamilton (2015, 2019), in which informative priors over the contemporaneous structural matrix combine with the likelihood to deliver set-identified parameters whose posteriors remain asymptotically responsive to prior information. The approach is well suited to natural-gas inference for two reasons. First, short-run supply and demand elasticities in gas markets are notoriously difficult to estimate from the data alone, since supply and demand shocks generate observationally similar correlations between price and quantity. Second, the alternative of purely agnostic sign restrictions provides too little discipline in a market where the relevant elasticities are small in absolute value. A related Bayesian framework has been applied to the United States natural gas market by Rubaszek

et al. (2021) and Farag (2024); this paper is, to our knowledge, the first application to the European market with separated pipeline and LNG supply channels. The dataset is constructed from JODI-Gas reports covering 22 EU member states between January 2010 and the most recent month of complete coverage, complemented by Dutch TTF prices from the World Bank Commodity Price Database and the euro-area industrial production index from Eurostat.

Three findings stand out from the posterior estimates and the structural analysis. First, the two supply channels display sharply different short-run elasticities. The posterior median for pipeline-and-domestic supply is $\gamma_{qp} \approx 0.08$, near-rigid, while LNG supply is substantially more responsive at $\beta_{qp} \approx 1.6$, consistent with the literature documenting LNG as the flexible balancing margin in Europe. Second, the impulse-response analysis shows that pipeline supply shocks transmit to prices as transient spikes that largely dissipate within five months, whereas LNG supply shocks transmit more persistently by shifting the medium-run market-clearing condition. Third, the forecast-error variance decomposition assigns roughly comparable short-horizon shares to supply and demand shocks—in contrast to much of the oil-market literature, in which demand-driven shocks dominate—and identifies storage-related precautionary demand as a significant amplification channel during episodes of supply stress.

We exploit the identified structural model to construct three counterfactual scenarios for the 2022 energy crisis. Holding LNG imports frozen at their February 2022 level for the duration of the crisis window produces a median price gap of approximately +0.80 log-points by the end of the sample: prices would have been roughly 2.2 times the realised level. Allowing LNG to fully absorb the contemporaneous pipeline-and-domestic shortfall, abstracting from Europe’s regasification bottleneck, produces a gap of -1.70 log-points (prices would have been about 18% of the realised level). Eliminating the autonomous demand-reduction signal identified by the SVAR over the formal EU policy window of Council Regulation 2022/1369 produces a gap of +0.30 log-points. The realised price path lies closer to the LNG-frozen upper envelope than to the LNG-unconstrained lower envelope, an asymmetry consistent with the regasification bottleneck having bound during the crisis: cargoes were globally redirectable, but Europe’s capacity to receive and distribute them was operationally constrained.

A fourth, forward-looking exercise extends the same framework to the 2026 Strait of Hormuz crisis, which is partially captured by the estimation sample. A fan of three scenarios for the path of European LNG over April 2026 to March 2027 yields a sharply asymmetric price response: under prompt resolution, prices revert to the baseline projection within a few months (-0.05 log-points by March 2027, or roughly 5% below the unconstrained projection), while under persistent disruption the gap widens to +0.30 log-points ($\approx +34\%$), and under an escalating second-wave path to +0.82 log-points ($\approx +127\%$, or $2.3\times$ the baseline). Public-domain evidence available at the time of writing places the realised May-2026 trajectory between the persistent and escalating scenarios, considerably closer to the former. The fan highlights geographic diversification of LNG supply, alongside receiving-side infrastructure, as a complementary energy-security instrument against the kind of chokepoint risk that Qatar’s exposure to the Strait of Hormuz exemplifies.

The remainder of the paper is organised as follows. Section 2 reviews the related literature on energy-market SVARs and on the contractual and infrastructural differences between pipeline gas and LNG. Section 3 presents the econometric methodology, including the prior, likelihood and posterior of the Baumeister–Hamilton framework. Section 4 describes the data, the empirical specification of the contemporaneous structure, and the priors imposed on the elasticities. Section 5 reports the posterior estimates, impulse responses, forecast-error variance decompositions, historical decomposition, and the three counterfactual scenarios, together with a synthesis of their joint reading. Section 6 concludes.

2 Literature Review

This chapter relates the present study to three strands of literature. First, it reviews the use of Structural Vector Autoregressive (SVAR) models in the crude oil market, which provides the conceptual and methodological foundation for most empirical work on energy price shocks. Second, it surveys the growing literature applying SVARs to natural gas markets, with primary emphasis on Europe, complemented by evidence from the United States and, to a lesser extent, Asia. Finally, it reviews the literature on contractual and infrastructural differences between pipeline gas and liquefied natural gas (LNG), which motivates the explicit separation of supply sources in the empirical model.

The modern empirical analysis of energy price shocks originates in the crude oil literature. Seminal work by Hamilton (1996) documented a strong negative relationship between oil price increases and U.S. macroeconomic activity, giving rise to an extensive literature on the identification and transmission of oil price shocks. A key methodological breakthrough was introduced by Kilian (2009), who proposed a Structural VAR framework distinguishing between oil supply shocks, global aggregate demand shocks, and oil-specific demand shocks related to precautionary or speculative motives. This approach emphasized that oil price increases can have markedly different macroeconomic effects depending on their underlying source. Subsequent contributions by Kilian and Murphy (2012) and Kilian and Murphy (2014) highlighted limitations of purely agnostic sign restrictions and argued for incorporating economically meaningful restrictions on short-run elasticities and inventory behavior. Building on these insights, Baumeister and Hamilton (2015) and Baumeister and Hamilton (2019) developed a Bayesian SVAR framework under set identification, showing that informative priors on structural elasticities can discipline the identification problem without imposing exact zero restrictions. Applied to the global oil market, this methodology revealed that demand-driven shocks account for the bulk of oil price fluctuations, while supply shocks play a more limited role.

Three lessons from this literature are directly relevant for natural-gas inference: separating supply- and demand-driven shocks is essential for understanding price dynamics; short-run elasticities of energy supply and demand are typically very small, so that even moderate shocks generate large price movements; and credible identification often requires informative priors, sign restrictions or auxiliary observables such as inventories.

In contrast to crude oil, European natural gas markets have historically been regionally fragmented owing to transportation constraints and infrastructure specificity. European supply

has long relied on pipeline imports under rigid long-term contracts, with LNG playing a limited role until recently, and early empirical work focused on the linkage between gas and oil prices under oil-indexed contracts; Szafranek and Rubaszek (2024) document that this relationship weakened markedly from the mid-2000s onward, reflecting market liberalisation and the transition to hub-based pricing. Recent SVAR analyses of the European market identify supply shocks, demand shocks related to economic activity or weather, and storage-driven precautionary demand, and typically find relatively inelastic short-run demand together with constrained supply responses. The 2022 energy crisis has sharpened the role of supply-side and substitution mechanisms within this framework. Güntner et al. (2024) show that the collapse in Russian pipeline deliveries represented a dominant negative supply shock partly offset by LNG inflows, with precautionary storage demand amplifying the price response. A central implication, emphasised also by Moll et al. (2023) and Hamilton (2023), is that aggregating pipeline gas and LNG into a single supply measure conceals large compositional adjustments and understates the role of supply-side shocks, motivating models that distinguish between supply sources explicitly.

The U.S. natural gas market provides a useful comparator. It is largely self-sufficient, with price formation dominated by domestic market-based production, and SVAR analyses there have focused on disentangling domestic supply shocks—in particular those associated with the shale-gas revolution—from demand shocks driven by weather, economic activity and storage behaviour. Rubaszek et al. (2021) estimate short-run demand elasticities larger in absolute value than those typically found for oil, suggesting that fuel-switching in electricity generation and industrial use renders U.S. gas demand more price-responsive than in Europe; they also document a non-negligible role for storage-related demand shocks. Farag (2024) highlight the growing international dimension of the U.S. market following the expansion of LNG export capacity: as the United States has become a major LNG exporter, domestic prices have become increasingly exposed to global demand conditions, though the magnitude of cross-regional transmission remains smaller than in more import-dependent regions.

A large body of research shows that pipeline gas and LNG represent fundamentally different supply margins. Pipeline gas is typically governed by long-term, route-specific contracts with take-or-pay and destination clauses, which imply low short-run price elasticity and limited flexibility in response to shocks Molnar (2022). LNG supply, by contrast, has become increasingly flexible as contract durations shorten, destination restrictions weaken, and spot market trading expands, allowing cargoes to be redirected across regions in response to price differentials Sharples (2023). This flexibility gives LNG a strategic role beyond its volume share, as it provides a credible outside option that constrains the market power of pipeline exporters Ritz (2019). Evidence from the 2022 European energy crisis illustrates these mechanisms clearly: rising LNG imports partially offset the collapse in Russian pipeline flows, though only at exceptionally high spot prices, underscoring both the substitutability and the distinct pricing dynamics of the two supply channels.

Taken together, this literature implies that pipeline gas and LNG differ markedly in contractual rigidity, cost structures, elasticity, and adjustment speed. Ignoring these differences risks mischaracterizing both the sources of natural gas price fluctuations and their macroeconomic consequences. This insight provides a central motivation for the empirical strategy

adopted in this chapter, which explicitly distinguishes between pipeline and LNG supplies within a Bayesian SVAR framework.

3 Methodology

The model can be rewritten in matrix form as a typical structural VAR (SVAR).

$$\mathbf{A}y_t = \mathbf{B}x_{t-1} + u_t \quad (1)$$

$$u_t \sim N(0_{n \times 1}, \mathbf{D}) \quad (2)$$

where $y_t = (y_{1t}, y_{2t}, \dots, y_{nt})'$ is the $n \times 1$ vector of endogenous variables, $\mathbf{A}$ is the $n \times n$ matrix that contains all the contemporaneous elasticities across the endogenous variables, $x_{t-1} = (y_{t-1}, \dots, y_{t-m})'$ is the $k \times 1$ vector of lagged regressor, $\mathbf{B}$ is the $n \times k$ autoregressive matrix of coefficients and $\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$ is the variance of the structural innovations.

But it also has a reduced form, which can be interpreted as a conventional VAR:

$$y_t = \Phi x_{t-1} + \varepsilon_t \quad (3)$$

$$\varepsilon_t \sim N(0_{n \times 1}, \Omega) \quad (4)$$

$$\Phi = A^{-1}B \quad (5)$$

$$\Omega = A^{-1}D(A^{-1})' \quad (6)$$

The model is estimated using Bayesian methods with sign restrictions. In set-identified models, the prior of the contemporaneous elasticities does not vanish asymptotically, as Baumeister and Hamilton (2015) (henceforth BH) demonstrated. The methodology was subsequently extended and applied to the global crude oil market by Baumeister and Hamilton (2019) and to the U.S. natural gas market by Rubaszek et al. (2021) and Farag (2024). The BH framework offers the flexibility to center all priors of the elements in $\mathbf{A}$ at values supported by the literature. Anchoring the location and scale of the prior for the structural elasticities at empirically grounded values lets the model incorporate information that is not available from the data alone. In this section we provide a brief description of the methodology.

3.1 Prior

The starting point for setting the prior is decomposing $p(A, B, D)$ as:

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D}) = p(\mathbf{A}) \times p(\mathbf{D} \mid \mathbf{A}) \times p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}) \quad (7)$$

The prior for the diagonal variance-covariance matrix $\mathbf{D}$ consists in setting that the inverse of its elements d_{ii}^{-1} are jointly independent and follow a Gamma distribution with $\kappa_i, \tau_i(\mathbf{A})$ as parameters.

$$p(\mathbf{D} \mid \mathbf{A}) = \prod_{i=1}^n p(d_{ii} \mid \mathbf{A}) \quad (8)$$

$$d_{ii}^{-1} \mid \mathbf{A} \sim \Gamma(\kappa_i, \tau_i(\mathbf{A})), \quad (9)$$

Similarly, rows of matrix $\mathbf{B}$ are assumed to be jointly independent and to follow a multivariate normal distribution with parameters m_i, M_i .

$$p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}) = \prod_{i=1}^n p(b_i \mid \mathbf{D}, \mathbf{A}) \quad (10)$$

$$b_i \mid \mathbf{A}, \mathbf{D} \sim N(m_i, d_{ii}M_i), \quad (11)$$

The prior for $\mathbf{A}$ can be freely chosen to reflect any knowledge about the structural relations between the endogenous variables. In the case of this study it will be the joint distribution of several truncated Student-t.

3.2 Likelihood

Innovation of the structural model is assumed to be normal.

$$p(Y_T \mid \mathbf{A}, \mathbf{D}, \mathbf{B}) = (2\pi)^{-Tn/2} |\det(\mathbf{A})|^T \det(\mathbf{D})^{-T/2} \quad (12)$$

$$\times \exp \left\{ -\frac{1}{2} \sum_{t=1}^T (\mathbf{A}y_t - \mathbf{B}x_{t-1})' \mathbf{D}^{-1} (\mathbf{A}y_t - \mathbf{B}x_{t-1}) \right\}, \quad (13)$$

3.3 Posterior

To proceed with the posterior distribution, we need a similar decomposition as before:

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D} \mid Y_t) = p(\mathbf{A} \mid Y_t) \times p(\mathbf{D} \mid \mathbf{A}, Y_t) \times p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}, Y_t) \quad (14)$$

where the conditional posterior of the elements of the diagonal variance–covariance matrix, $p(\mathbf{D} \mid \mathbf{A}, Y_t)$, is expressed as the joint distribution of all elements on the diagonal:

$$p(\mathbf{D} \mid \mathbf{A}Y_t) = \prod_{i=1}^n p(d_{ii} \mid \mathbf{A}Y_t) \quad (15)$$

$$d_{ii}^{-1} \mid \mathbf{A} \sim \Gamma(\kappa_i^*, \tau_i^*(\mathbf{A})), \quad (16)$$

The parameters κ_i^* and $\tau_i^*(\mathbf{A})$ that fully characterize this posterior are functions of the data and of $\mathbf{A}$, as follows:

$$\kappa_i^* = \kappa_i + T/2 \quad (17)$$

$$\tau_i^*(\mathbf{A}) = \tau_i(\mathbf{A}) + \xi_i^*(\mathbf{A}) \quad (18)$$

And the term ξ_i^* refers to the sum of squared residuals from regressing $\tilde{Y}_i(\mathbf{A})$ on $\tilde{X}_i$ defined as:

$$\xi_i^*(\mathbf{A}) = (\tilde{Y}_i(\mathbf{A})' \tilde{Y}_i(\mathbf{A})) - (\tilde{Y}_i(\mathbf{A})' \tilde{X}_i) (\tilde{X}_i' \tilde{X}_i)^{-1} (\tilde{X}_i' \tilde{Y}_i(\mathbf{A})) \quad (19)$$

$$\tilde{Y}_i = [y_1' a_i \quad \cdots \quad y_T' a_i \quad m_i' P_i]'_{(T+k) \times 1} \quad (20)$$

$$\tilde{X}_i = [x_0' \quad \cdots \quad x_{T-1}' \quad P_i]'_{(T+k) \times k} \quad (21)$$

where $M_i = P_i' P_i$ and a_i are the rows of the matrix $\mathbf{A}$.

The conditional posterior of the autoregressive matrix, $p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}, Y_t)$, can be defined as the joint posterior of its rows:

$$p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}, Y_t) = \prod_{i=1}^n p(b_i \mid \mathbf{D}, \mathbf{A}) \quad (22)$$

$$b_i \mid \mathbf{A}, \mathbf{D} Y_t \sim N(m_i^*, d_{ii} M_i^*) \quad (23)$$

where

$$m_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1} (\tilde{X}_i' \tilde{Y}_i), \quad (24)$$

$$M_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1}. \quad (25)$$

Finally, the posterior of $\mathbf{A}$ is:

$$p(\mathbf{A} \mid \mathbf{Y}_T) = \frac{k_T p(\mathbf{A}) [\det(\mathbf{A} \hat{\Omega}_T \mathbf{A}')]^{T/2}}{\prod_{i=1}^n [(2\tau_i^*/T)^{\kappa_i^*}]} \prod_{i=1}^n \left\{ \frac{|\mathbf{M}_i^*|^{1/2}}{|\mathbf{M}_i|^{1/2}} \frac{\tau_i^{\kappa_i}}{\Gamma(\kappa_i)} \Gamma(\kappa_i^*) \right\}.$$

where $\hat{\Omega}_T = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_t \hat{\varepsilon}_t'$ is the sample variance covariance matrix estimated by maximum likelihood.

Inference draws from $p(\mathbf{A} \mid \mathbf{Y}_T)$ using a Random-Walk Metropolis–Hastings sampler. The proposal density is a multivariate Gaussian with covariance $\xi \cdot \hat{\mathcal{H}}^{-1}$, where $\hat{\mathcal{H}}$ is the Hessian of the log-posterior evaluated at the mode and the scaling factor is fixed at $\xi = 0.0625$. We run 2×10^6 draws with a burn-in of 1.7×10^6 , retaining the remaining 3×10^5 post-burn-in draws for inference. Cumulative impulse responses, forecast-error variance decompositions, historical decompositions, and the counterfactual scenarios in Section 5.5 are computed for each retained draw and summarised by their pointwise posterior median and 5–95% quantiles. The random seed is fixed at 140778 for reproducibility. Convergence is assessed by visual inspection of the trace plots of the contemporaneous elasticities and by comparing the prior–posterior marginals reported in Section 5.1.

4 Empirical model and data

4.1 Data

We model the European natural gas market as a five-variable system that splits supply into two channels with markedly different short-run flexibility. The first channel, denoted q_{pip} , comprises net pipeline imports plus EU domestic production. Both share contractual and physical rigidities that prevent rapid adjustment within a month. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay clauses, fixed transport infrastructure and limited spare capacity (Molnar, 2022). Domestic production is constrained by reservoir decline curves and regulatory frameworks that operate over multi-year horizons; the gradual phase-out of the Groningen field illustrates how the latter, rather than monthly price

dynamics, drives variation in this margin. We aggregate the two because they share the relevant economic property for the question at hand—near-zero short-run price elasticity—which makes them the slow-adjusting margin of European gas supply.

The second channel, q_{LNG} , comprises net LNG imports. LNG operates as the flexible margin of European supply: cargoes are redirectable across regions in response to price differentials, contract durations have shortened, destination clauses have weakened, and exposure to spot prices has expanded (Sharples, 2023; Ritz, 2019). The 2022 energy crisis illustrates this asymmetry sharply: in response to the collapse in Russian pipeline deliveries, EU LNG imports surged by over 60% within months, while pipeline-and-domestic supply showed virtually no compensating expansion in the aggregate.

We deliberately keep q_{pip} as a single aggregate rather than introducing domestic production as a third VAR variable. The substantive distinction the model identifies is rigid versus flexible supply, and that distinction is economically binary: domestic production behaves like pipeline imports (inelastic, slow-adjusting) and not like LNG. Adding a third supply variable would consume identification room without delivering economically distinct shocks, and the resulting series would be dominated by the Groningen-related structural decline rather than by short-run market dynamics. The robustness of this aggregation choice is verified in Appendix I, where the model is re-estimated with q_{pip} replaced by the IEA Gas Trade Flows pipeline aggregate (external producer-country flows only, no domestic production).

The dataset comprises five variables at monthly frequency. Pipeline-and-domestic supply q_{pip} and LNG imports q_{LNG} are constructed from the JODI-Gas database, summing reported pipeline imports, LNG imports and domestic production across 22 EU member states.¹ Total natural gas inventories s are also taken from JODI; the variable is included because of its central role in consumption smoothing and in precautionary or speculative behaviour. Industrial production ip is the manufacturing index for the euro area, sourced from Eurostat. The real price of natural gas p is the Dutch TTF benchmark from the World Bank Commodity Price Database, deflated by the euro area HICP.

$$y_t = [q_{\text{pip}}, ip, p, s, q_{\text{LNG}}] \quad (26)$$

All series are expressed in logarithmic growth rates (in percentage points per month) to remove unit roots in prices, industrial production and quantities. The sample spans February 2010 through the most recent month with complete JODI coverage, March 2026.² The pipeline, LNG, and inventory series are seasonally adjusted with X-13 ARIMA-SEATS, industrial production is sourced from Eurostat already seasonally adjusted, and the real gas

¹Country panel: AT, BE, CZ, DE, DK, EE, ES, FI, FR, GR, HU, IE, IT, LT, LU, LV, NL, PL, PT, SE, SI, SK. Five EU members are excluded due to incomplete JODI coverage over 2010–2024: BG, HR, MT and RO have data only from 2019 onwards, and CY does not report. The 22 countries jointly account for approximately 95% of EU-27 natural gas consumption. For the inventory series (see below), NL is additionally dropped because both its closing-stock-level and stock-change reports start only in January 2014; Germany’s pre-2013 inventory level is reconstructed by backward-extrapolating from January 2013 using monthly aggregate stock changes, which DE reports throughout the sample. A robustness check using IEA *Gas Trade Flows* data in place of JODI for the two supply aggregates is reported in Appendix I.

²March 2026 is also the first full month of the Hormuz crisis. A small number of observations at the end of the sample are preliminary; the treatment of preliminary releases and the reconstruction of monthly stock changes for four small countries from closing stock levels are detailed in Appendix A.

price is left in raw form (not seasonally adjusted). The VAR is estimated with $p = 6$ monthly lags, which is the longest specification permitted by the available degrees of freedom and is consistent with the lag choice in Baumeister and Hamilton (2019) and Rubaszek et al. (2021); the impulse responses are reported up to a horizon of 24 months.

4.2 Specification and assumptions

This section describes the structural model used in the paper to make the identification for the European natural gas market. The main characteristics of the contemporaneous elasticities among endogenous variables are equivalent to those used in Baumeister and Hamilton (2019); Rubaszek et al. (2021); Farag (2024).

We begin with the structural equations of the two supply components. Both pipeline and LNG imports are assumed to respond contemporaneously only to innovations in the gas price. Consistent with previous studies, both supply types are considered to be mildly elastic; that is, the corresponding supply price elasticities γ_{qp} and β_{qp} are assumed to differ from zero. Because the production and delivery of natural gas involve adjustment costs, imports are expected to react to real activity shocks only with a delay of at least one month. Moreover, since the output of these two technologies is extracted simultaneously, they cannot be adjusted relative to one another within the same period.

$$q_t^{pip} = \gamma_{qp}p_t + \mathbf{b}'_1x_{t-1} + \varepsilon_t^{\text{pipeline supply}} \quad (27)$$

$$q_t^{LNG} = \beta_{qp}p_t + \mathbf{b}'_5x_{t-1} + \varepsilon_t^{\text{LNG supply}} \quad (28)$$

Real activity is broadly proxied by industrial production, which is assumed to respond contemporaneously only to the gas price. This assumption is motivated by the fact that a supply disruption affects industrial activity primarily through its impact on prices. In the short run, however, a shortfall in gas supply can be partially offset by substituting other components of the energy mix, such as biogas or alternative electricity sources. Nevertheless, any increase in the gas price is expected to exert a negative effect on industrial production.

$$ip_t = \alpha_{yp}p_t + \mathbf{b}'_2x_{t-1} + \varepsilon_t^{\text{aggregate demand}} \quad (29)$$

Demand for natural gas is separated into *flow demand* and *precautionary demand*. Flow demand captures shifts in the portion of gas that is effectively consumed, and can be associated with changes in market conditions that make gas more desirable to consumers. The structural demand equation incorporates two contemporaneous channels: industrial production affects gas demand through the income elasticity $\alpha_{qy} > 0$, and the gas price affects demand through the price elasticity $\alpha_{qp} < 0$. Since consumers are indifferent to the source of natural gas supply, any unit of pipeline gas, LNG, or stored gas is treated as identical, so aggregate demand is defined as $Q_t = q_t^{pip} + q_t^{LNG} - s_t$. In this case, the structural demand equation is given by:

$$Q_t = \alpha_{qy}ip_t + \alpha_{qp}p_t + \mathbf{b}'_3x_{t-1} + \varepsilon_t^{\text{flow demand}}, \quad (30)$$

or equivalently, in inverse demand form:

$$\alpha_{qp}p_t = q_t^{pip} + q_t^{LNG} - s_t - \alpha_{qy}ip_t - \mathbf{b}'_3x_{t-1} - \varepsilon_t^{\text{flow demand}}. \quad (31)$$

Table 1: Contemporaneous elasticities in $\mathbf{A}$

Parameter	Meaning	Sign restriction
γ_{qp}	Pipeline supply elasticity	$\gamma_{qp} > 0$
β_{qp}	LNG supply elasticity	$\beta_{qp} > 0$
α_{yp}	Effect of p on economic activity	$\alpha_{yp} < 0$
α_{qy}	Income elasticity of gas demand	$\alpha_{qy} > 0$
α_{qp}	Gas demand elasticity	$\alpha_{qp} < 0$
ψ_1	Effect of q^{pip} on gas inventories	none
ψ_2	Effect of p on gas inventories	none
ψ_3	Effect of q^{LNG} on gas inventories	none

Precautionary demand captures changes in market participants' expectations that lead to adjustments in natural gas inventories. Such behavior may arise either because agents anticipate a potential supply disruption or because a rise in prices triggers speculative motives among inventory holders. Stocks of gas are adjusted in response to current imports, making inventories a potential substitute for them. Inventories also react to price innovations within the same month, because any shock that raises the gas price has two competing effects on stocks. On one hand, a higher price gives inventory holders the opportunity to realize a profit by selling at a price above their acquisition cost, implying that stocks decline as the price rises ($\psi_2 < 0$). On the other hand, if inventory holders interpret the current price increase as a precursor of further price hikes, they may prefer to expand current inventories in order to sell more gas in the future ($\psi_2 > 0$). These two conflicting forces make the sign of the price elasticity ψ_2 theoretically ambiguous, and thus, like the other two inventory parameters ψ_1 and ψ_3 , it is left without a sign restriction.

$$s_t = \psi_1 q_t^{pip} + \psi_2 p_t + \psi_3 q_t^{LNG} + \mathbf{b}'_4 x_{t-1} + \varepsilon_t^{\text{precautionary demand}} \quad (32)$$

4.3 Matrix form

Following the specification of a SVAR the model is rewritten in matrix form as it follows:

$$\begin{pmatrix} 1 & 0 & -\gamma_{qp} & 0 & 0 \\ 0 & 1 & -\alpha_{yp} & 0 & 0 \\ 1 & -\alpha_{qy} & -\alpha_{qp} & -1 & 1 \\ -\psi_1 & 0 & -\psi_2 & 1 & -\psi_3 \\ 0 & 0 & -\beta_{qp} & 0 & 1 \end{pmatrix} \begin{pmatrix} q_t^{pip} \\ ip_t \\ p_t \\ s_t \\ q_t^{LNG} \end{pmatrix} = \mathbf{B}x_{t-1} + \begin{pmatrix} u_t^{\text{pipeline supply}} \\ u_t^{\text{aggregate demand}} \\ u_t^{\text{flow demand}} \\ u_t^{\text{precautionary shock}} \\ u_t^{\text{LNG supply}} \end{pmatrix}$$

Where the structural shocks are assumed to be normally distributed $u_t \sim N(0_{n \times 1}, \mathbf{D})$ and the contemporaneous elasticities in $\mathbf{A}$ satisfy the following five sign restrictions:

$$\gamma_{qp} > 0 \quad \beta_{qp} > 0 \quad \alpha_{yp} < 0 \quad \alpha_{qy} > 0 \quad \alpha_{qp} < 0$$

The way to implement these restrictions is to incorporate them in the prior of the contemporaneous matrix, $p(\mathbf{A})$, assigning truncated distributions in each case. The purpose of this is twofold, first it makes easier to incorporate prior knowledge of the elements in $\mathbf{A}$ that is not directly provided by the likelihood of the data. On top of that it helps to ensure that

the posterior of the elasticities $p(\mathbf{A} \mid Y_t)$ gives most of the probability to values of $\mathbf{A}$ that make the impact matrix $\mathbf{A}^{-1}$ to satisfy the following signs.

$$\begin{pmatrix} \varepsilon_{q^{pip},t} \\ \varepsilon_{ip,t} \\ \varepsilon_{p,t} \\ \varepsilon_{s,t} \\ \varepsilon_{q^{LNG},t} \end{pmatrix} = \begin{pmatrix} + & + & + & ? & ? \\ + & + & - & ? & + \\ - & + & + & + & - \\ ? & ? & + & ? & ? \\ ? & + & + & ? & + \end{pmatrix} \begin{pmatrix} u_t^{\text{pipeline supply}} \\ u_t^{\text{aggregate demand}} \\ u_t^{\text{flow demand}} \\ u_t^{\text{precautionary shock}} \\ u_t^{\text{LNG supply}} \end{pmatrix}$$

4.4 Prior knowledge about these elasticities

The structural elasticities embedded in matrix A , which govern the contemporaneous interactions between quantities, prices, and real economic activity, can be disciplined using a well-established empirical literature on energy markets, particularly natural gas and oil. In set-identified SVAR models that rely on inequality or sign restrictions, these elasticities play a central role in shaping inference. Unlike in fully identified models, the posterior distribution does not converge to the likelihood even asymptotically, implying that prior information does not wash out with increasing sample size (Baumeister and Hamilton, 2015). As a consequence, the choice of prior distributions is not innocuous and must be grounded in economically and empirically plausible values.

The short-run price elasticity of pipeline natural gas supply, denoted by γ_{qp} , measures the responsiveness of pipeline gas imports to price changes. We impose the theoretical restriction $\gamma_{qp} > 0$ and center the prior at 0.1, reflecting the well-established view that natural gas supply is highly inelastic in the short run. Empirical studies consistently find small elasticities, typically close to zero or below 0.1, due to technological constraints, adjustment costs, and contractual rigidities (Al-Sahlawi, 1989; Dahl and Duggan, 1996; Arora, 2014; Rubaszek et al., 2021; Mason and Roberts, 2018). Centering the prior at a small positive value therefore captures an upward-sloping but steep supply curve, while the heavy-tailed Student- t prior allows for uncertainty without admitting implausibly large elasticities.

A similar prior is imposed on the short-run price elasticity of LNG supply, β_{qp} . Although LNG-specific elasticity estimates are scarce, short-run LNG supply is constrained by fixed liquefaction capacity, shipping availability, and regasification infrastructure, implying only limited responsiveness to price signals. We therefore impose $\beta_{qp} > 0$ and center the prior at 0.1, allowing the data to assess whether LNG supply is more flexible than pipeline supply while anchoring inference in a plausible range.

On the demand side, the short-run price elasticity of natural gas consumption, α_{qp} , is expected to be negative. We center the prior at -0.3 and truncate the distribution to enforce $\alpha_{qp} < 0$. This calibration reflects the broad consensus that gas demand is highly inelastic in the short run. Early evidence summarized by Al-Sahlawi (1989) documents short-run elasticities clustered around values close to -0.3 , while more recent studies typically estimate elasticities between -0.04 and -0.16 (Joutz, 2009; Bernstein and Madlener, 2011; Arora, 2014), with demand remaining far from elastic even when allowing for heterogeneity (Burke and Yang, 2016).

The elasticity of gas demand with respect to aggregate economic activity, α_{qy} , captures the income elasticity of natural gas demand. We impose $\alpha_{qy} > 0$ and center the prior at 0.7, reflecting a strong but less than proportional response of gas consumption to changes in output. This calibration is motivated by survey evidence summarized by Al-Sahlawi (1989), which reports income elasticities ranging from near zero to above unity, as well as by more recent cross-country estimates for OECD economies that cluster around 0.7 Burke and Yang (2016). Anchoring the prior at this benchmark provides an empirically grounded characterization of demand responsiveness to economic activity.

Finally, the parameter α_{yp} captures the contemporaneous response of aggregate economic activity to changes in natural gas prices. We impose a negative sign restriction and center the prior at -0.05 , reflecting the well-established view that higher energy prices act as a drag on output by increasing production costs and reducing real disposable income, while having only a modest immediate effect Kilian (2008). Similar calibrations are standard in Bayesian SVAR studies of energy markets, and posterior estimates are typically close to zero Baumeister and Hamilton (2019). Evidence from the European gas crisis likewise points to noticeable but non-catastrophic short-run output effects Güntner et al. (2024), supporting a small negative prior mean.

In summary, all prior distributions are centered on values firmly supported by the empirical literature. Supply elasticities are assumed to be small and positive, demand price elasticities moderate and negative, income elasticities positive and sizable, and the contemporaneous output response to gas prices modestly negative. By adopting literature-based priors within the Bayesian framework of Baumeister and Hamilton (2015), the model achieves economically meaningful identification while avoiding improper posteriors. Although the data remain free to update these beliefs, the priors provide a necessary anchor that ensures sensible inference in a weakly identified SVAR setting.

5 Results

In this section, we present the estimation results of the baseline model and discuss the dynamics of the European natural gas market. We begin by examining the posterior distributions of the model parameters, focusing in particular on the short-run price elasticities of demand and supply. Then, we analyze the posterior impulse response functions, which capture the dynamic responses of the endogenous variables to structural shocks. Next, we quantify both the short- and long-term effects of these shocks on the variables included in the VAR system and assess their historical decompositions, highlighting their contributions to the variation in natural gas prices. Finally, we conduct a counterfactual analysis based on a hypothetical scenario in which European LNG imports were not available during the post-energy-crisis recovery period.

5.1 Posterior knowledge of the elasticities

We begin by comparing the prior and posterior distributions of the parameters in the contemporaneous relations matrix $\mathbf{A}$, with particular attention to the posterior medians of the

Table 2: Priors and posteriors distributions for parameters in **A**

	γ_{qp}	β_{qp}	α_{yp}	α_{qy}	α_{qp}	ψ_1	ψ_2	ψ_3
<i>Priors affecting contemporaneous coefficients A</i>								
Location	0.1	0.1	-0.05	0.7	-0.3	0	0	0
Scale	0.2	0.2	0.1	0.3	0.1	0.5	0.5	0.5
Degrees of freedom	3	3	3	3	3	3	3	3
Sign restriction	t^+	t^+	t^-	t^+	t^-	t	t	t
<i>Posterior of contemporaneous coefficients A</i>								
5%	0.034	0.825	-0.028	0.446	-1.098	-0.157	-0.108	-0.011
50%	0.079	1.629	-0.009	0.894	-0.530	-0.041	-0.067	0.030
95%	0.125	3.028	-0.001	1.535	-0.283	0.073	-0.026	0.072
Mean	0.079	1.745	-0.011	0.929	-0.592	-0.042	-0.067	0.030

short-run demand and supply elasticities. Figure 1 displays the prior distributions (red line) alongside the posterior ones (blue histograms), while the corresponding descriptive statistics are reported in the lower panel of Table 2. The main findings can be summarized as follows.

The posterior median of the short-run price elasticity of pipeline-and-domestic supply, γ_{qp} , is estimated at only 0.079, indicating a highly inelastic response of the rigid supply margin to price changes. This estimate is broadly consistent with the elasticities reported by Rubaszek et al. (2021) and Farag (2024) for the U.S. natural gas market, who find elasticities of approximately 0.01 and 0.02, respectively, and conclude that American gas supply is strongly price inelastic. It is also of the same order of magnitude as the elasticity estimated by Mason and Roberts (2018) for natural gas wells in Wyoming, who report a value of 0.026. Moreover, our estimate is comparable to the European supply elasticities obtained by Adolfsen et al. (2024) and to those reported by Güntner et al. (2024) for the German natural gas market, further confirming the limited short-run responsiveness of contracted-and-domestic gas supply to price movements.

The price elasticity of LNG supply, β_{qp} , tells a sharply different story. Its posterior median is 1.63, with a 90% credible interval of [0.83, 3.03] that excludes any value below unity. LNG supply is therefore estimated to be highly responsive to price: a one-percent increase in the real gas price is associated with a contemporaneous expansion of LNG imports of roughly 1.6%. The contrast with $\gamma_{qp} \approx 0.08$ is striking—one full order of magnitude—and provides the clearest empirical support for the rigid-versus-flexible characterisation that motivates the distinction between the two supply channels. The estimate is consistent with the institutional view of LNG as the marginal balancing source in Europe: cargoes can be redirected across regions in response to price differentials, contract durations have shortened, and exposure to spot pricing has expanded (Sharples, 2023; Ritz, 2019).

Regarding demand elasticities, the posterior median of the short-run price elasticity, α_{qp} , is estimated at -0.53 , with a 90% credible interval of $[-1.10, -0.28]$. In absolute terms this value is moderately larger than the estimates reported by Rubaszek et al. (2021) and Farag (2024) for the U.S. natural gas market (-0.42 and -0.18 , respectively). The result suggests that European demand is somewhat more responsive to price than its U.S. counterpart, a finding consistent with the larger role of fuel-switching in electricity generation and with the greater exposure of European industrial users to spot pricing rather than long-term take-or-pay contracts. The estimate is also broadly in line with the upper end of the cross-country range surveyed by Burke and Yang (2016).

As for the income elasticity, α_{qy} , its posterior distribution remains close to the prior, with a median of 0.89 , broadly consistent with the 0.719 reported by Güntner et al. (2024) for the German market and the 0.725 obtained by Baumeister and Hamilton (2019) for crude oil. This similarity indicates that shifts in aggregate economic activity translate into comparable changes in demand across both energy commodities and across different markets.

For the remaining parameters, the estimates indicate that, in the short run, the effect of natural gas prices on economic activity, α_{yp} , is almost negligible, with a posterior median of -0.009 . This value is well below both our prior belief and the estimates reported by Adolfsen et al. (2024) for the euro area, yet it is close to the -0.002 obtained by Baumeister and Hamilton (2019) for the crude oil market.

Turning to the inventory equation, the contemporaneous effects of pipeline-and-domestic supply (ψ_1), prices (ψ_2), and LNG imports (ψ_3) on inventories show modest magnitudes but distinct patterns. The posterior median of ψ_2 is -0.067 , with a 90% credible interval entirely below zero ($[-0.108, -0.026]$): a higher contemporaneous price reduces inventories, consistent with stockholders releasing gas to capture the price gain rather than accumulating inventories in anticipation of further increases. The posterior median of ψ_1 is small and not significantly different from zero (-0.041 , $[-0.157, 0.073]$). By contrast, ψ_3 has a posterior median of $+0.030$, with a 90% credible interval that almost entirely lies above zero ($[-0.011, 0.072]$): positive innovations to LNG imports translate contemporaneously into higher inventories. The opposite signs of ψ_1 and ψ_3 are economically intuitive in light of the institutional differences between supply channels: rigid (pipeline-and-domestic) gas is largely delivered directly into the system, whereas LNG cargoes are typically buffered in regasification and storage facilities before distribution.

5.2 Impulse Response Functions

We decompose the dynamics of the European natural gas market into five structural shocks, following the structural oil market SVAR literature Kilian (2009); Kilian and Murphy (2012, 2014); Baumeister and Hamilton (2019). The model distinguishes two supply shocks, pipeline gas and LNG, and three demand shocks, aggregate demand, flow demand, and precautionary or storage demand. We trace their cumulative impulse responses for the set of endogenous variables.

First, consider the IRFs to a pipeline supply shock. An adverse pipeline shock lowers pipeline imports on impact and raises the real natural gas price, while industrial production declines

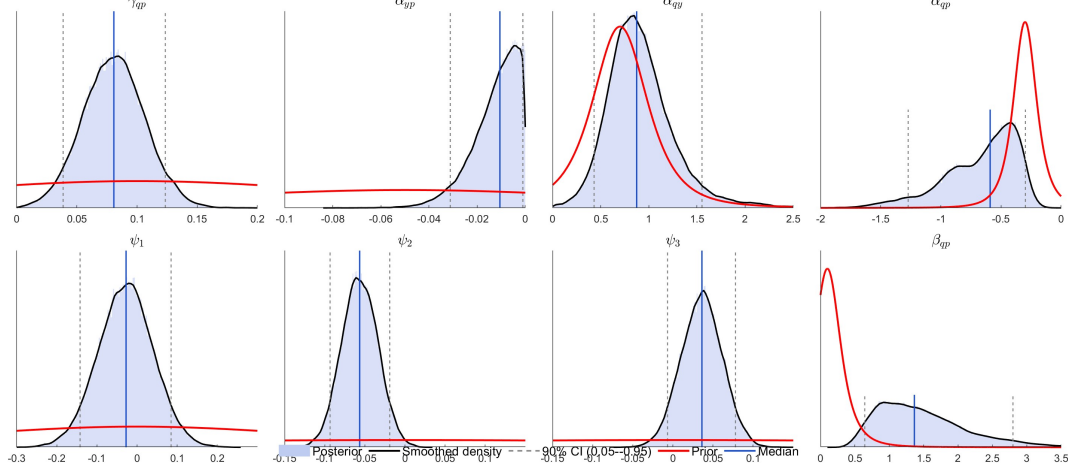


Figure 1: Prior and posterior of contemporaneous elasticities for the baseline model. Each panel reports one element of the contemporaneous matrix $\mathbf{A}$: γ_{qp} (pipeline supply elasticity), α_{yp} (industrial-activity response to price), α_{qy} (demand income elasticity), α_{qp} (demand price elasticity), ψ_1 , ψ_2 and ψ_3 (storage equation, capturing the effects of pipeline supply, price, and LNG supply on inventories), and β_{qp} (LNG supply elasticity). The light-blue filled histogram shows the marginal posterior; the solid black line is the corresponding smoothed (kernel) density; the vertical dashed grey lines mark the 5% and 95% posterior quantiles (90% credible interval); the solid vertical blue line marks the posterior median; the solid red line is the prior density used for the parameter (see the prior specification in Section 4.4 and the legend at the bottom).

in the short run. Inventories fall for a few months, consistent with an initial drawdown that partially cushions the physical shortfall, and LNG imports increase as supply is reallocated toward alternative inflows. In our estimates, the price effect is transitory and dissipates within about five months after the shock.

Next, consider the IRFs to a positive LNG supply shock that raises total gas availability, leading to a sustained decline in the real natural gas price and a persistent improvement in market conditions. Inventories increase for an extended period as the additional supply supports storage accumulation, while the adjustment in other inflows is gradual. Conversely, an adverse LNG shock reduces LNG availability, generates a prolonged tightening of the market, and raises prices in a persistent manner, reflecting the central role of LNG as the medium-run balancing margin in Europe.

These dynamics align with the gas literature documenting temporary price effects of pipeline disruptions. For example, Adolfsen et al. (2024) also find that pipeline supply shocks raise natural gas prices only temporarily, although their price response persists for up to nine months. A natural explanation is that their supply shock aggregates pipeline and LNG supply, so the estimated response reflects a mixture of short-lived pipeline disruptions and more persistent LNG-related adjustment. By distinguishing the two supply channels, our framework isolates the shorter-lived nature of pipeline shocks more clearly. The European experience in 2022, marked by the sharp reduction in Russian pipeline deliveries, provides

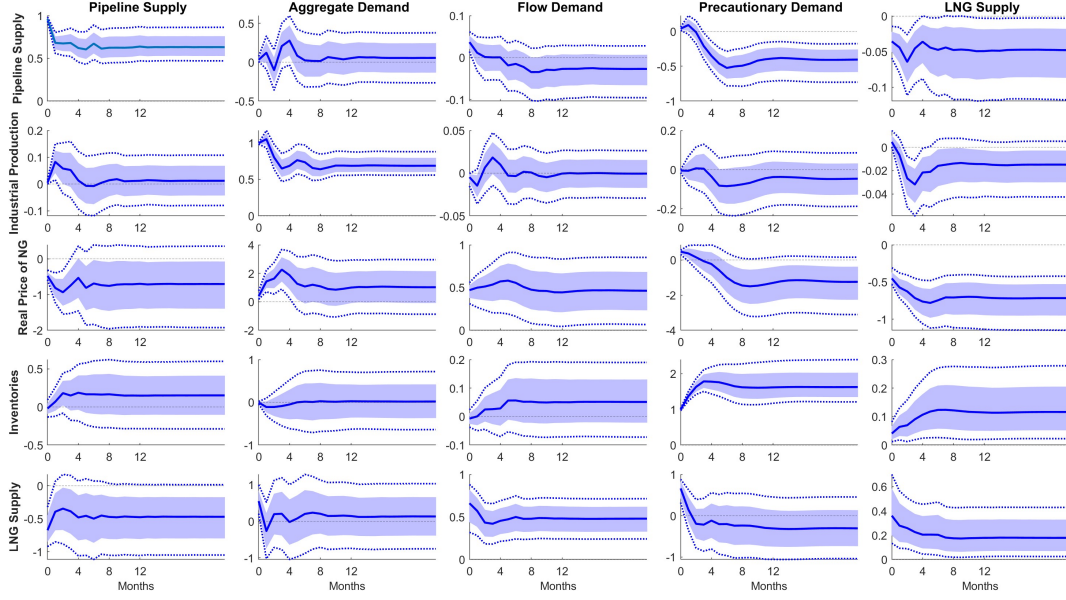


Figure 2: Cumulative Impulse Responses of the endogenous variables (rows) to the five structural shocks (columns). The solid blue line is the posterior median, the light-blue shaded area is the 68% credible band, and the dotted blue lines are the 90% credible band. The horizon is 12 months. All responses are cumulated. Row labels denote the responding variable; column titles denote the identified shock.

a salient illustration of this mechanism, with pronounced price increases accompanied by comparatively limited immediate output losses as inventories and substitute imports partially mitigated the supply shortfall.

Compared with pipeline shocks, LNG shocks tend to be more persistent because LNG is the marginal balancing source in Europe and therefore shapes the medium-run market-clearing price. A pipeline disruption is typically a corridor-specific shortfall whose price impact is partly offset through inventory withdrawals, demand adjustment, and substitution toward other inflows, so the price effect can fade relatively quickly. An LNG supply shock instead reflects a change in Europe’s access to flexible global cargoes, which relaxes the constraint on replacement supply and shifts the marginal supply schedule for longer. As a result, LNG shocks affect prices more durably than pipeline shocks in the impulse responses.

Turning to aggregate and flow demand shocks (second and third columns), both generate upward pressure on natural gas prices, but they differ in the persistence of the price response. Aggregate demand shocks raise prices only temporarily, consistent with a gradual adjustment of quantities as the economy reverts toward baseline. Flow demand shocks, in contrast, trigger a sharper contemporaneous tightening of market conditions, reflected in an immediate price increase and a pronounced drawdown of inventories, and they may be mildly contractionary for industrial production as higher energy costs and short-run scarcity constrain activity. Over time, however, the price effect of flow demand shocks fades as consumption normalizes and inventories are replenished.

Finally, precautionary demand shocks capture shifts in desired inventory holdings driven by

Table 3: Median FEVD of the Price

	Pipeline Sup.	Aggr. Dem.	Flow Dem.	Prec. Dem.	LNG Sup.
1 month	22.1	18.1	20.3	20.3	19.1
2 months	16.9	53.8	8.6	12.1	8.6
3 months	17.3	54.1	7.6	13.5	7.5
6 months	15.6	61.4	4.5	14.0	4.5
1 year	14.8	59.6	3.9	17.9	3.9
2 years	14.7	59.5	3.9	18.1	3.8

Notes: Median forecast error variance decomposition (in percent) of the natural gas real price at horizons of 1, 2, 3, 6, 12 and 24 months. Rows do not necessarily sum to 100 because the table reports posterior medians, not posterior shares of a single draw. Data are monthly, 2010–2024.

expectations of future supply scarcity. A positive precautionary shock induces a simultaneous increase in inventories and natural gas prices, consistent with stockpiling behavior. In response, LNG imports remain broadly unchanged, whereas pipeline imports decline and settle at a permanently lower level after about three months. This pattern is suggestive of forward-looking inventory management, whereby storage holders anticipate persistent disruptions in pipeline supply, as in the 2022 episodes, and adjust their sourcing and inventory decisions accordingly.

Overall, the cumulative impulse responses indicate that European natural gas prices reflect the interaction of demand conditions, supply rigidities, and storage behavior. In the short run, price fluctuations are driven by both supply- and demand-side shocks, whereas medium-run dynamics are mostly associated with flow demand and LNG supply adjustments. These patterns are consistent with evidence from the oil and gas literatures and highlight the particular sensitivity of gas prices to infrastructure constraints and concerns about supply security. The horizon-dependent structure also carries a direct implication for monetary policy. Because pipeline-supply shocks transmit to gas prices as relatively short-lived spikes, they are likely to translate into transient headline-inflation deviations that a central bank can credibly look through; LNG-supply shocks, by contrast, propagate more persistently and may therefore exert pressure on medium-run inflation expectations and on core measures through second-round effects, consistent with the gas-to-inflation pass-through documented for the euro area by Adolfsen et al. (2024).

5.3 Variance Decomposition

A Forecast Error Variance Decomposition (FEVD) quantifies the relative importance of each structural shock in explaining the uncertainty of gas price fluctuations over time, thereby providing a summary measure of the dominant drivers of price variability. Table 3 reports the median forecast error variance decomposition (FEVD) of the natural gas price at different horizons.

At very short horizons, price fluctuations are driven by a relatively balanced mix of all the shocks in the model. At the one-month horizon, the contributions range from 18.1 percent

for aggregate demand to 22.1 percent for pipeline supply, with flow demand, precautionary demand and LNG supply each accounting for roughly 20 percent. This pattern reflects the fact that in the immediate aftermath of shocks, gas prices react to a wide range of disturbances affecting both supply and demand conditions. As the horizon increases, aggregate demand shocks emerge as the dominant source of price variance. From two months onward, aggregate demand explains more than 50 percent of the forecast error variance, rising above 60 percent at the six-month horizon. This indicates that medium- to long-run movements in natural gas prices are strongly linked to broader macroeconomic conditions and demand forces, rather than to short-lived supply shocks.

Turning to the relative roles of supply and demand shocks, the variance decomposition reveals clear differences in their persistence and horizon dependence. On the demand side, flow demand shocks, which capture short-run adjustments in industrial and residential consumption, play an important role at very short horizons, accounting for more than 20 percent of price variance at the one-month horizon. Their contribution, however, declines rapidly and falls below 5 percent beyond one year, indicating that immediate consumption adjustments have largely transitory effects on price volatility. In contrast, precautionary demand shocks display a more persistent pattern. While their short-run contribution is moderate, their importance increases with the forecast horizon, explaining close to 20 percent of price variance at one- and two-year horizons. Unlike flow demand shocks, which dissipate once short-run consumption adjusts, precautionary motives affect prices through expectations and inventory decisions that operate over longer horizons. This suggests that uncertainty-driven behavior and expectations about future supply conditions generate lasting effects on prices, in line with the evidence from the energy crisis.

On the supply side, both pipeline and LNG supply shocks contribute more modestly to overall price variance when averaged over the full sample. Pipeline supply shocks explain a non-negligible share of short-run price variability but stabilize at around 15 percent at longer horizons, pointing to a limited yet persistent influence. Meanwhile, LNG supply shocks account for an even smaller fraction of price variance, falling to below 5 percent at horizons beyond six months. This indicates that while supply disruptions, particularly in LNG markets, can trigger sharp price movements during specific episodes, their contribution to overall price uncertainty is relatively limited in the long run.

Taken together, these results highlight a clear horizon-dependent structure in which short-run price fluctuations reflect a combination of supply disturbances and immediate demand adjustments, whereas medium- and long-run volatility is increasingly shaped by aggregate demand conditions and persistent precautionary shocks rather than by supply-side forces.

5.4 Historical Decomposition

A historical decomposition expresses the evolution of an observed time series as the cumulative contribution of the structural shocks identified in the model. In a structural VAR framework, this tool is particularly useful for attributing large price movements to distinct economic mechanisms and for assessing how the relative importance of supply, demand, and precautionary forces varies over time. Rather than focusing on the response to isolated

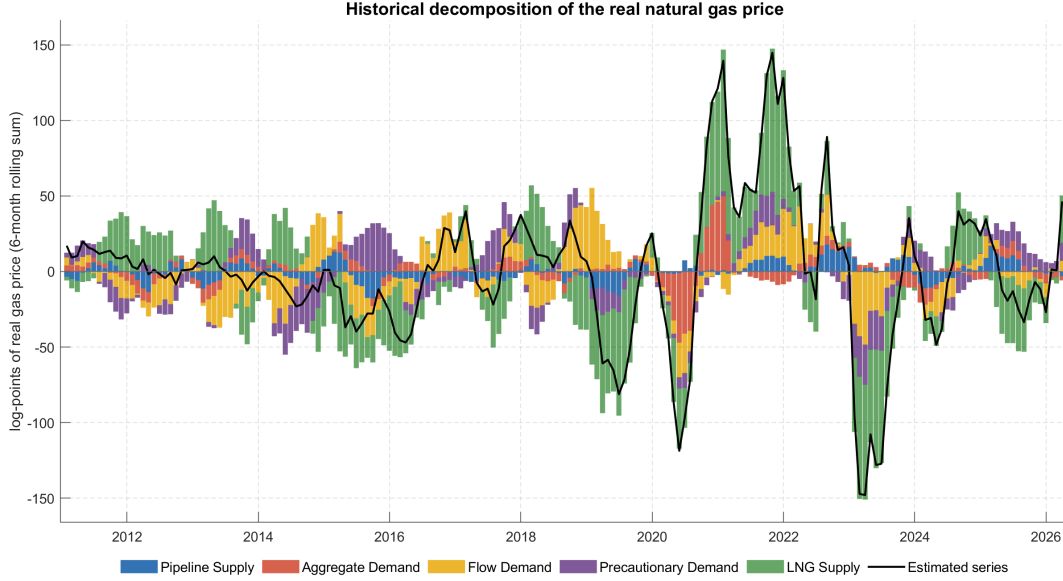


Figure 3: Historical decomposition of the European real natural gas price. Stacked coloured bars show the cumulative six-month contribution of each of the five identified structural shocks (steel blue: pipeline supply; terracotta: aggregate demand; mustard: flow demand; muted purple: precautionary/storage demand; sage green: LNG supply). The solid black line is the model-implied estimated series (sum of all shock contributions). Values are in log-points of the real gas price, plotted at monthly frequency over the period 2010–2024.

shocks, the historical decomposition provides a narrative interpretation of realized outcomes, allowing one to identify which shocks were most relevant during specific historical episodes.

The dynamics of European natural gas real prices are best understood as the outcome of three interacting forces: (i) very low short-run demand elasticity, (ii) binding supply and infrastructure constraints, and (iii) the pivotal role of storage and expectations when supply security deteriorates. Because gas is hard to substitute in heating and many industrial processes in the short run, even moderate physical imbalances translate into large price movements. In Europe, these imbalances are magnified by the fact that pipeline supply is tied to specific corridors and contracts, while LNG is globally traded but constrained by liquefaction capacity, shipping availability, and regasification infrastructure. Hence, price spikes emerge when the market must simultaneously (a) replace lost pipeline volumes, and (b) rebuild inventories ahead of winter, especially under heightened uncertainty. Figure 3 presents the historical decomposition of the European real natural gas price based on the five structural shocks identified in the model. The black line depicts the estimated log-level price, while the colored bars show the contemporaneous contribution of pipeline supply, aggregate demand, flow demand, precautionary demand, and LNG supply shocks. Several episodes align closely with well-documented market developments. Following the 2014 Crimea invasion and the deterioration in EU Russia energy relations, the decomposition suggests an increase in storage-related and flow pressures, although price effects remained contained, consistent with the gradual nature of the adjustment and the buffering role of contracts and infrastructure. The COVID-19 shock in 2020 appears as a demand-driven

episode: the sharp decline in prices coincides with large negative contributions from flow demand and precautionary demand, consistent with the collapse in economic activity and energy use documented for global gas markets during the pandemic International Energy Agency (2020). In the post-COVID recovery of 2021, prices start rising well before the full-scale 2022 crisis. This pattern is consistent with the combination of a rapid rebound in demand and supply-side frictions in global LNG markets and shipping ³, together with unusually low European storage entering the winter season Popkostova (2022). The most pronounced regime shift occurs during the 2022 energy crisis, when the sudden reduction in Russian pipeline deliveries forced Europe to rely on scarce flexible LNG volumes under tight global capacity conditions International Energy Agency (2023). In this period, the decomposition is dominated by positive contributions from flow demand, precautionary demand, and LNG supply shocks, consistent with scarcity concerns and storage motives playing a central role in price formation. Aggregate demand contributes comparatively little during the spike, which is also consistent with contemporaneous assessments that emphasize supply insecurity and risk premia rather than macroeconomic overheating Furtuna et al. (2022). Finally, the normalization in 2023 coincides with the unwinding of these forces and with improved fundamentals, including strong storage buffers and the gradual expansion of LNG import capacity and market resilience EuropeanCommission (2024).

Overall, the historical decomposition clarifies how the same structural mechanisms operate with very different intensity across episodes. Periods of stability are characterized by offsetting shocks that keep prices anchored, whereas crisis episodes arise when multiple supply- and risk-related forces reinforce each other. The analysis shows that large price movements emerge not from a single dominant shock, but from the interaction between constrained supply, expectations, and limited adjustment margins. This perspective helps rationalize both the severity of the 2022 price spike and the speed of its subsequent reversal.

5.5 Counterfactual Analysis

The structural VAR framework employed in this paper enables a rich set of counterfactual and scenario-based analyses, in line with Güntner et al. (2024) and Antolín-Díaz et al. (2021). We construct three different counterfactual scenarios to evaluate the role of LNG and demand adjustment in shaping the price trajectory during the 2022–2023 European energy crisis. Table 4 summarises the headline results before each scenario is discussed in turn.

We follow the convention of Antolín-Díaz et al. (2021) and construct counterfactuals at the level of the observed endogenous variables (a *data* counterfactual) rather than by zeroing out subsets of structural shocks. Concretely, for each counterfactual scenario we impose an alternative path on one of the endogenous variables over the crisis window, re-run the historical decomposition on the modified dataset, and reconstruct the implied path of the real gas price by accumulating the resulting monthly log-growth rates from January 2022. This approach is well suited to the questions we ask: each scenario is naturally framed in

³Specially a stronger power and gas demand in Asia pulled LNG away from Europe Reuters (2021). This has combined with a global supply crunch and limited injections, resulting in record-low stocks in the second half of 2021

Table 4: Summary of the three counterfactual scenarios

Scenario	Constraint	Window	Median gap	90% CI	Ratio
CF1 LNG frozen at Feb-22	$g_t^{LNG} = 0$	Feb-22 – Dec-23	+0.80	[0, +2.3]	$\times 2.23$
CF2 LNG absorbs pipeline shortfall	$\Delta q_t^{LNG} = -\Delta q_t^{pip}$	Feb-22 – Dec-23	-1.70	[-5, -0.5]	$\times 0.183$
CF3 No autonomous demand reduction	$u_t^3 = 0$	Aug-22 – Mar-23	+0.30	[-0.05, +0.85]	$\times 1.35$

Notes: Median price gap is the posterior median of $\log(p_t^{CF}) - \log(p_t)$ evaluated at the end of the sample, expressed in log-points; the 90% credible interval is the corresponding [5, 95] posterior range. Ratio is $\exp(\text{median gap})$ and reports the implied counterfactual level relative to the observed series (e.g. a ratio of $\times 2.20$ means the counterfactual price is 2.2 times the observed one). Counterfactuals 1 and 2 are framed as polar bounds on what the LNG margin could have delivered (upper and lower bounds respectively); Counterfactual 3 measures the joint contribution of the autonomous demand reduction over the formal EU policy window.

terms of a counterfactual time path for an observed margin of adjustment (LNG imports, flow demand) rather than in terms of an isolated structural innovation. The first scenario asks how prices would have evolved had LNG imports not been able to expand above their pre-crisis level; the second asks how prices would have responded under a more aggressive demand-saving programme; and the third asks how prices would have evolved if LNG had fully absorbed the loss of Russian pipeline supply.

5.5.1 First Counterfactual: LNG cannot adapt during the crisis

In response to the looming gas crisis in 2022, European policymakers moved swiftly to diversify gas supply by expanding imports of liquefied natural gas (LNG). Germany, which had relied entirely on pipeline gas until then, fast-tracked the construction of floating LNG terminals, the first of which became operational in December 2022. The EU also negotiated deals with alternative suppliers (for example, a joint EU–US task force secured additional LNG deliveries of at least 15 bcm for 2022) and maximised use of existing import facilities. As a result, Europe’s LNG imports surged by over 60% in 2022, adding roughly 66 bcm to supply, primarily from the United States, Qatar and other partners.

Our first counterfactual asks what would have happened if this expansion had not been possible. We hold the level of net LNG imports constant at its February 2022 value throughout the crisis window (February 2022–December 2023); the remaining variables of the VAR (pipeline-and-domestic supply, industrial production, the real price, and inventories) are allowed to adjust endogenously according to the estimated structural relationships. This is an extreme version of the scenario, since it shuts down both the policy-driven push and the price-driven expansion of LNG, and therefore should be read as an *upper bound* on the contribution of the LNG response to price stabilisation.⁴

Figure 4 reports the full effect of the counterfactual on the three endogenous variables of the SVAR in which LNG plays a meaningful role: the real natural gas price, industrial

⁴A less extreme variant, in which LNG continues to grow at its pre-crisis trend rather than being completely frozen, is reported in Appendix E. Under that milder specification the median price gap is only +0.04 log-points, suggesting that most of the observed LNG expansion during 2022–2023 was the continuation of a strong pre-crisis trend rather than a crisis-induced policy response.

production, and gas inventories. Each row corresponds to one variable; the left column reports the log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red) with the 5–95% posterior credible band; the right column reports the median posterior gap (counterfactual minus baseline) with the same credible band, expressed in log-points of the corresponding level.

The headline result is in the top row. The median price gap stabilises at approximately +0.8 log-points from late 2023 onwards, with a 90% credible interval that broadly spans $[0, +2.3]$. In level terms, the point estimate implies that the real gas price would have been roughly $\exp(0.8) - 1 \approx 1.2$ times higher than its observed (and baseline) level, or about 2.2 times the realised price, had LNG imports remained frozen at their February 2022 volume throughout the crisis window. The credible band is wide and tilted upward, reflecting both posterior uncertainty about the structural elasticities and the fact that the SVAR is linear: it does not capture the demand-destruction and policy-response mechanisms that would dampen prices below the model-implied path at extreme valuations. The point estimate should therefore be read as a qualitative upper bound for the marginal contribution of the post-February 2022 LNG expansion to price stabilisation, not as a literal forecast of where prices would have settled.

The effects on the other two endogenous variables affected by the LNG-path constraint, gas inventories and industrial production, are of opposite sign and very different magnitudes. The inventory gap is the most clearly identified secondary effect: it settles around -0.13 log-points by 2024 (a counterfactual stock level roughly 12% below baseline) with a 90% credible interval whose upper bound never exceeds +0.05 log-points. This is consistent with the model’s contemporaneous structure: when LNG cargoes do not arrive, inventories cannot be replenished ($\psi_3 > 0$), and the higher prices simultaneously trigger sales of stored gas ($\psi_2 < 0$). The industrial-production gap, by contrast, is faintly positive (+0.015 log-points, roughly +1.5% in levels) with a 90% credible interval that comfortably straddles zero. The contemporaneous price-on-output elasticity $\alpha_{yp} \approx -0.009$ is essentially nil, so the direct channel through which high prices would depress activity is not strongly identified, and the impulse response of IP to an LNG-supply shock is itself small and slightly negative (plausibly capturing the positive co-movement between unexpected LNG arrivals and the kind of global recessions that redirect cargoes to Europe rather than Asia). The SVAR is unable to attribute meaningful real-activity consequences to the LNG margin in this counterfactual, and the symmetric exercise in Counterfactual 2 yields a similarly muted IP response.

5.5.2 Second Counterfactual: LNG fully absorbs the pipeline shortfall

Our second counterfactual mirrors the first but reverses the direction of the LNG path constraint. We ask what prices would have been if LNG imports had not merely expanded but had fully compensated for the loss of Russian pipeline supply during the crisis. Concretely, we construct a deterministic counterfactual LNG *level* path anchored at the February 2022 LNG volume and rising one-for-one with the contemporaneous shortfall in pipeline-and-domestic supply:

$$\text{LNG}^{cf}(t) = \text{LNG}_{\text{obs}}(\text{Feb-22}) + [q_{\text{obs}}^{pip}(\text{Feb-22}) - q_{\text{obs}}^{pip}(t)], \quad t \in [\text{Feb-22}, \text{Dec-23}].$$

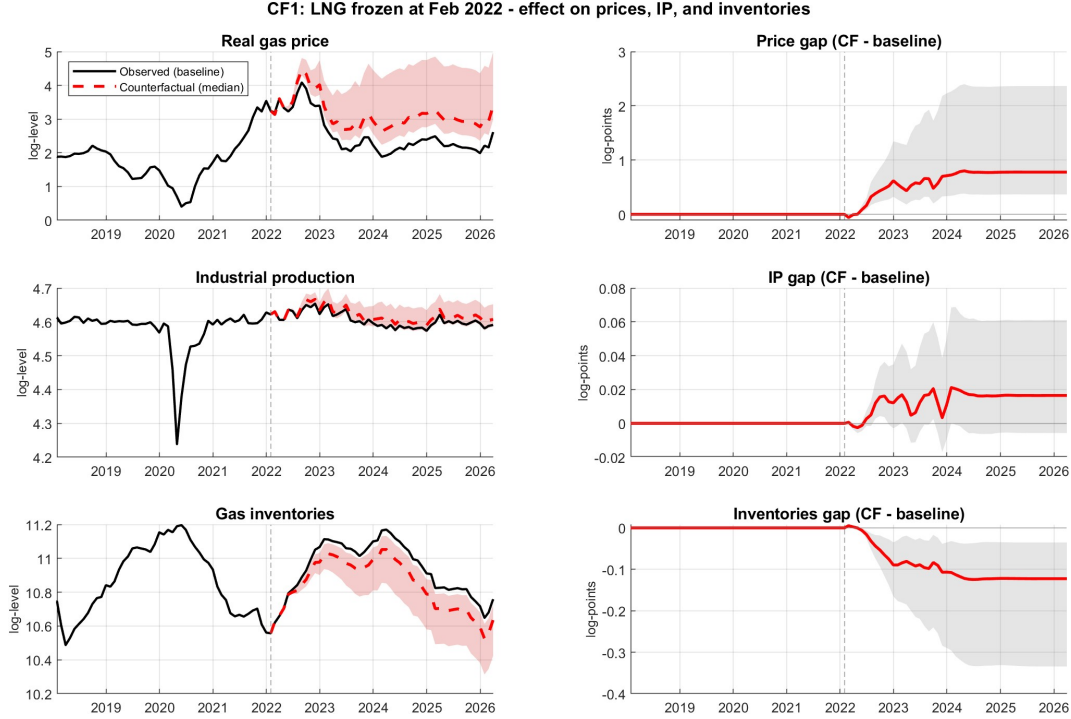


Figure 4: Counterfactual 1: full multi-variable response if LNG imports had been frozen at their February 2022 level until December 2023. Each row corresponds to one of the three endogenous variables affected by the LNG-path constraint (real gas price, industrial production, gas inventories). *Left column:* log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red), with the 5–95% credible band for the counterfactual. *Right column:* median posterior gap (counterfactual minus baseline) with the same credible band, in log-points. The counterfactual is implemented as a forward simulation in which the monthly growth rate of LNG imports is forced to zero during the crisis window and the LNG-supply structural innovation absorbs the constraint (Antolín-Díaz et al., 2021); other structural innovations are kept at their historical values. The baseline reconstruction coincides with the observed series by construction.

The resulting series is then converted to monthly log-growth rates and imposed on the SVAR using the same forward simulation procedure as in Counterfactual 1, with the LNG-supply structural innovation absorbing the constraint and all other historical innovations kept at their estimated values (Antolín-Díaz et al., 2021). The Feb-22 anchor is deliberate: anchoring instead at observed LNG would layer an additional pipeline-shortfall absorption on top of the substitution response already endogenously embedded in the observed LNG path, imposing a shock of size $(1 + \gamma)$ times the pipeline shortfall rather than 1 times, where γ is the implicit substitution coefficient that the SVAR identifies in the data. The Feb-22 anchor delivers a clean $1 \times$ shortfall, in symmetry with Counterfactual 1 (which freezes LNG at the same Feb-22 level).

This is again a deliberately extreme scenario, the symmetric counterpart of the first counterfactual: rather than asking what happens when the LNG margin is unavailable, it asks

what happens when the LNG margin is perfectly elastic. The imposed path implies a near-doubling of monthly LNG imports relative to their February 2022 level during the worst of the crisis (see Figure 10 in the appendix). Such volumes would have required regasification, downstream pipeline, and global cargo availability well above what was physically feasible in 2022–2023, precisely the bottleneck that the scenario assumes away. Together with Counterfactual 1, this exercise therefore brackets the contribution of LNG to the realised price path: the two scenarios delineate, respectively, an upper and a lower envelope for what the LNG margin could have delivered absent any physical constraint.

Figure 5 reports the full effect of the counterfactual on the three endogenous variables in which LNG plays a role. As in Figure 4, each row corresponds to one variable; the left column reports the log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red) with the 5–95% credible band; the right column reports the median posterior gap (counterfactual minus baseline), in log-points.

The headline result is again in the top row. The median price gap stabilises at approximately -1.7 log-points by 2024, with a 90% credible interval that broadly spans $[-5, -0.5]$. In level terms, the point estimate implies that the real gas price would have been about $\exp(-1.7) \approx 0.18$ times its observed level, or around 18% of it, had LNG fully absorbed the pipeline shortfall; equivalently, the realised price was roughly $\exp(1.7) \approx 5.5$ times the price the bottleneck-free counterfactual would have produced. Read against the realised peak of the Dutch TTF benchmark of roughly 340 EUR/MWh, the median counterfactual implies a price of around 60 EUR/MWh, only modestly above the pre-crisis “normal” range of 20–30 EUR/MWh. In words, the model is saying that a frictionless LNG response would have left the gas market only mildly perturbed by the Russian pipeline shock. The credible band is wide and tilted to the negative side, reflecting two compounding sources of uncertainty: (i) posterior dispersion in the structural elasticities (notably α_{qp} and β_{qp}) and (ii) the size of the imposed shock, a near-doubling of monthly LNG imports relative to their Feb-22 level, which pushes the SVAR outside the range of variation observed in the sample. Just as the magnitude reported for Counterfactual 1 should be read as a qualitative upper bound, this estimate is best understood as a qualitative *lower bound* on what prices could have been: it captures the direction and the substantial latent slack that an unconstrained LNG margin would have provided, not a literal forecast.

The remaining two rows of Figure 5 report the effects on inventories and industrial production. The two findings are essentially the mirror image of the results discussed for Counterfactual 1. The median inventory gap settles around $+0.25$ log-points by 2024, implying counterfactual stock levels roughly 28% above baseline, with a 90% credible interval that is almost entirely positive. As in Counterfactual 1, this is the most clearly identified secondary effect of the LNG margin: when cargoes arrive in abundance, storage is replenished aggressively ($\psi_3 > 0$) and the lower price level discourages drawdown ($\psi_2 < 0$). The median industrial-production gap is approximately -0.04 log-points, with a 90% credible interval that comfortably straddles zero. The interpretation is the same as in Counterfactual 1: the contemporaneous price-on-output elasticity $\alpha_{yp} \approx -0.009$ is essentially nil under the posterior, and the lagged transmission of the LNG-supply shock to IP is weak, so the SVAR cannot attribute a clearly identified real-activity consequence to the LNG margin within the

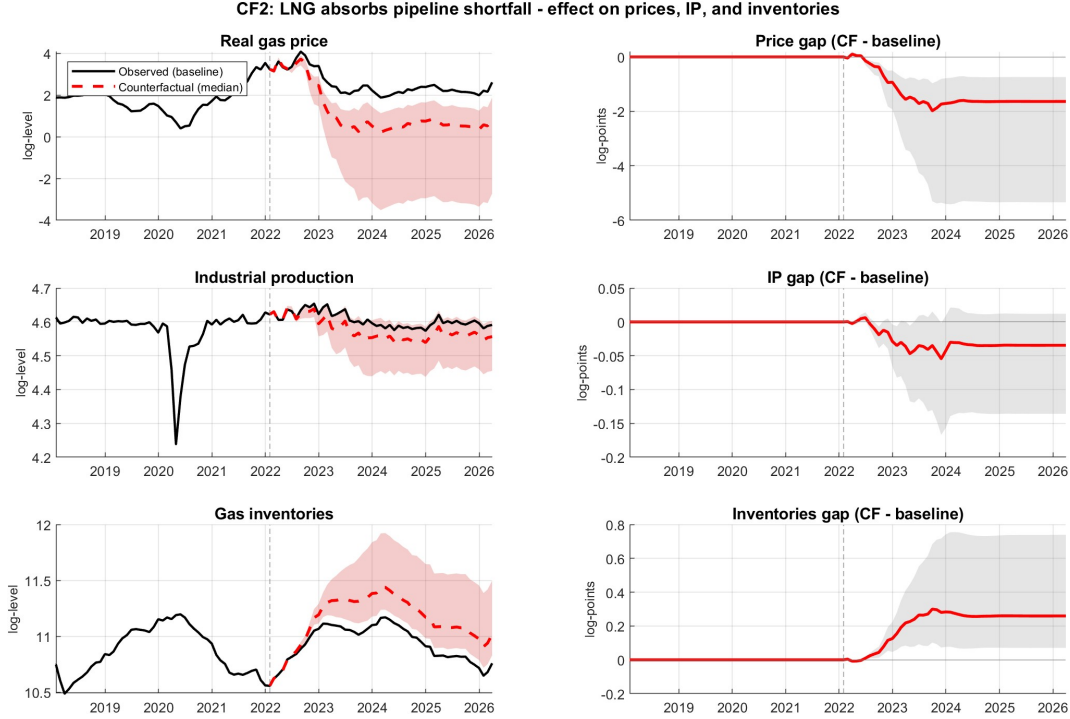


Figure 5: Counterfactual 2: full multi-variable response if LNG imports had fully absorbed the pipeline shortfall between February 2022 and December 2023. Each row corresponds to one of the three endogenous variables affected by the LNG-path constraint (real gas price, industrial production, gas inventories). *Left column:* log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red), with the 5–95% credible band for the counterfactual. *Right column:* median posterior gap (counterfactual minus baseline) with the same credible band, in log-points. The counterfactual LNG level path is anchored at the Feb-22 LNG volume and rises one-for-one with the contemporaneous pipeline-and-domestic shortfall; it is then imposed on the SVAR as a forward simulation in which the LNG-supply structural innovation absorbs the constraint, while other structural innovations remain at their historical values. The baseline reconstruction coincides with the observed series by construction.

sample window, regardless of whether the imposed shock is to suppress or to amplify LNG flows.

5.5.3 Third Counterfactual: No autonomous demand reduction

In parallel with the supply-side responses studied in Counterfactuals 1 and 2, the European Union enacted an unprecedented coordinated programme to curb natural gas demand during the crisis. The legislative timeline is sharp. On 8 March 2022 the European Council announced the REPowerEU framework calling for a rapid phase-out of Russian energy imports; the Commission detailed the plan on 18 May 2022 (REPowerEU Plan, COM/2022/230). On 20 July 2022 the Commission proposed the Communication *Save Gas for a Safe Winter* together with the legislative act Council Regulation (EU) 2022/1369, which the Council

adopted on 5 August 2022 and which entered into force on 9 August 2022 (Consilium press release, 26 July 2022). The Regulation set a voluntary target for member states to reduce gas consumption by at least 15% relative to their five-year average over the period from 1 August 2022 to 31 March 2023. National implementing measures spanned public-awareness campaigns, thermostat caps and heating curtailment in public buildings, industrial fuel switching, accelerated electrification of heat, and, in several member states, formal rationing protocols for non-protected industrial consumers. The Regulation was prolonged by Council Regulation (EU) 2023/706 on 30 March 2023, extending the 15% target through 31 March 2024. Outturn during the original window was a $\sim 18\%$ reduction in gas consumption relative to the five-year baseline, exceeding the target (Institute for Energy Economics and Financial Analysis (IEEFA), 2024).

Because gas consumption is not an observable in our SVAR (the system records pipeline-and-domestic supply, LNG supply, industrial production, the real price, and inventories), a clean counterfactual on the demand-reduction programme cannot be implemented as a data counterfactual on a single observable variable, as in Counterfactuals 1 and 2. Instead, we exploit the SVAR’s own identification. Our variable ordering $y_t = [q^{pip}, ip, p, s, q^{LNG}]$ places the inverse-demand equation in the third row, so that the third structural innovation, u_t^3 , captures the autonomous component of gas demand, that is, the residual of the demand equation after netting out the contemporaneous effects of income (via α_{qy}) and price (via α_{qp}). Inspection of the historical decomposition (Section 5.4) confirms the empirical signature one would expect from the policy: u^3 takes systematically negative values from autumn 2022 through mid-2023, consistent with an autonomous downward shift in gas demand beyond what income and price elasticities can explain. Counterfactual 3 sets this autonomous demand reduction to zero during the formal policy window:

$$u_{cf}^3(t) = 0 \quad \text{for } t \in [\text{Aug-2022, Mar-2023}],$$

while all other structural innovations are kept at their historical values. The window matches Council Regulation 2022/1369 exactly. Because u^3 is white noise by construction in a correctly identified SVAR ($E[u_t^3] = 0$), the counterfactual reference is simply the unconditional mean (zero); no additional calibration is required.

The negative u^3 sequence identified by the SVAR during the policy window bundles three economically distinct demand-reduction channels, which the counterfactual is unable to disentangle individually. The first channel is the EU and member-state policy package itself, comprising mandatory thermostat caps, industrial fuel switching, accelerated electrification and public-awareness campaigns. The second is the unusually warm winter of 2022–23, in which mean European temperatures stood roughly 1.5°C above the historical norm (Copernicus Climate Change Service (C3S), 2023). The third is any residual demand destruction by high prices that exceeds the contemporaneous elasticity α_{qp} captured by the SVAR, encompassing industrial closures, behavioural conservation responses, and expectations-driven precaution. The counterfactual therefore measures the joint contribution of all three channels, and not the policy package in isolation. We are explicit about this caveat: the SVAR cannot, with a single demand-equation residual, attribute the observed autonomous demand reduction to its mutually-correlated drivers. The interpretation that follows is correspondingly conservative: Counterfactual 3 reports what prices, output and inventories would have

been had the joint autonomous demand reduction not occurred. The EU policy is one of the three contributing forces, and likely the largest.

Figure 6 reports the full multi-variable response. The headline median price gap (counterfactual minus baseline) is positive by construction: removing the autonomous demand reduction should make prices higher. The median price gap peaks at approximately +0.40 log-points within the policy window and stabilises around +0.30 log-points by the end of the sample, with a 90% credible interval that broadly spans $[-0.05, +0.85]$. In level terms, the point estimate implies that the realised price during and after the window would have been roughly $\exp(0.30) \approx 1.35$ times higher, or about 35% above the observed series, had the autonomous demand reduction not occurred. Read against a post-crisis “normalised” Dutch TTF price of around 30–50 EUR/MWh, this is a wedge of order 10–17 EUR/MWh.

The two remaining endogenous variables show essentially no response. The median IP gap oscillates within ± 0.005 log-points during the window and settles indistinguishably from zero thereafter; the 90% credible interval comfortably straddles zero throughout. As in Counterfactuals 1 and 2, the SVAR cannot attribute meaningful real-activity consequences to the demand-shock channel within the sample. The median inventory gap is similarly small (+0.03 log-points by the end of the sample, with a 90% credible interval that straddles zero); the sign is mildly counterintuitive, since more autonomous demand would naively imply more drawdown and lower stocks, but the magnitude is within the noise of the SVAR’s storage equation and reflects the small contemporaneous loading $A_{4,3}^{-1}$ of the demand shock on inventories.

The cumulative path of u^3 in Figure 9 reveals a feature that motivated us to consider a longer constraint window. The posterior median of cumulative u^3 peaks shortly before the policy window opens, falls sharply during the window, and continues to drift downward through the end of 2023, consistent with the formal prolongation of the regulation through Council Regulation 2023/706 (valid until 31 March 2024). We examine an extended Aug-22–Dec-23 window in Appendix F. The extended exercise produces a median price gap of similar magnitude to the baseline ($\approx +0.30$ log-points), confirming that the demand-reduction signal identified by Counterfactual 3 carries through the prolonged window without significant offsetting innovations. The empirical content of the demand-side channel is therefore robust to the precise choice of window: whether one ends in March 2023 or in December 2023, the cumulative price contribution of the autonomous demand reduction is of the same order of magnitude.

A direct comparison of the magnitudes of the three counterfactuals is informative but must be made with care. The demand-side counterfactual delivers a price gap roughly 40% the size of the LNG-frozen scenario (Counterfactual 1: +0.80 lp) and roughly 18% the size of the LNG-unconstrained scenario (Counterfactual 2: -1.70 lp, in absolute value). A direct ranking of policy importance from these numbers would be misleading, however, because the three counterfactuals impose shocks of very different sizes in standard-deviation terms: Counterfactual 1 freezes LNG for nearly two years (an extreme upward push), Counterfactual 2 nearly doubles LNG against its Feb-22 baseline (an extreme downward push), and Counterfactual 3 zeroes out roughly eight months of moderate demand innovations. What the comparison *does* support, conservatively, is that both adjustment channels, namely supply-

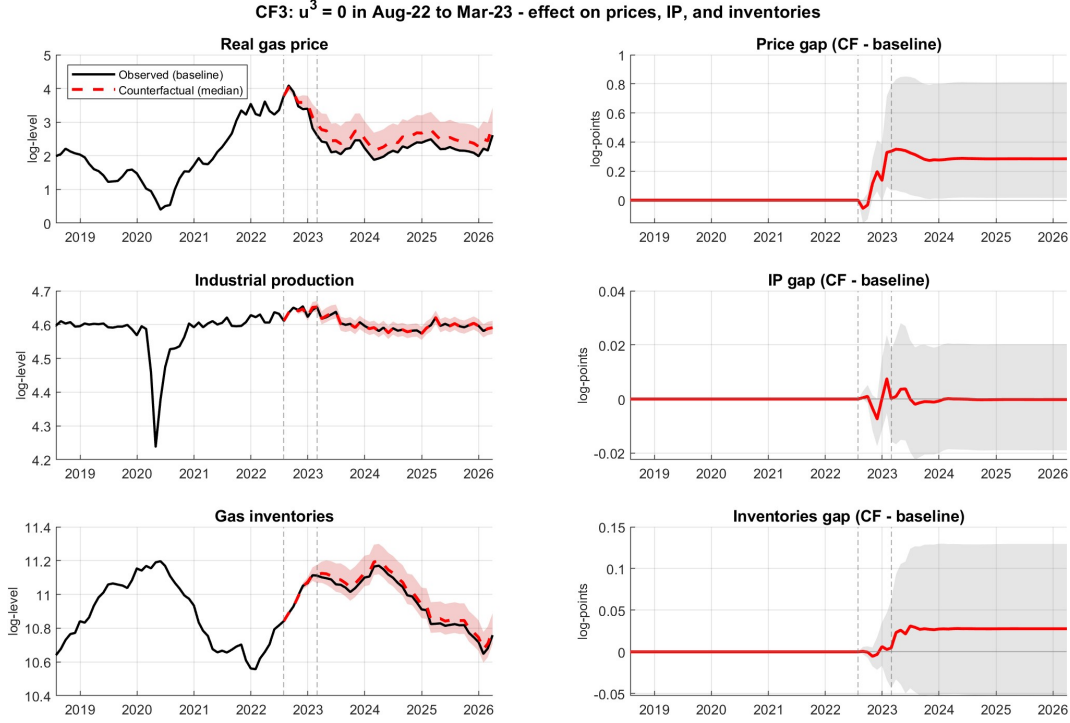


Figure 6: Counterfactual 3: full multi-variable response if the autonomous component of gas demand had been zero during the EU policy window (Aug-2022 to Mar-2023, Council Regulation 2022/1369). The third structural shock u^3 , identified by the SVAR as the residual of the inverse-demand equation, is set to zero during the window; all other structural innovations are held at their historical values. *Left column:* log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red), with the 5–95% credible band for the counterfactual. *Right column:* median posterior gap (counterfactual minus baseline) with the same credible band, in log-points. Two vertical dashed lines mark the start and end of the policy window. The baseline reconstruction coincides with the observed series by construction.

side LNG expansion and demand-side conservation, contributed materially and in the same direction to the realised stabilisation of European gas prices during the crisis. The supply side was quantitatively the larger contributor, but neither channel was negligible.

5.5.4 Bracketing the LNG margin and the regasification bottleneck

The three exercises are best read together. Counterfactuals 1 and 2 are deliberate polar cases of the LNG margin: the first shuts it off, the second makes it perfectly elastic. They yield price ratios of $\exp(0.80) \approx 2.23$ and $\exp(-1.70) \approx 0.18$ relative to the baseline. Counterfactual 3 operates on a different margin and produces a ratio of $\exp(0.30) \approx 1.35$, the price reduction attributable to the joint contribution of EU policy, the warm winter, and price-induced demand destruction over the formal Council Regulation 2022/1369 window. The realised series lies between the two LNG envelopes and decidedly closer to the upper one, which says that the observed LNG expansion captured only a fraction of the stabilisation that

an unconstrained LNG margin would have delivered. The wedge between Counterfactual 2 and the realised path is the cost of the constraint that prevented the rest of it: Europe’s limited regasification capacity, downstream pipeline take-away and storage-injection rate during 2022–2023.

The corresponding policy implication is concrete. Global LNG production was not the binding constraint in 2022. Spot cargoes were available; the EU–US task force secured volumes; Qatari and US producers were physically able to ship more. What constrained the European response was the receiving side: regasification terminals running near nameplate capacity, no FSRU in the Northwest European corridor before December 2022, downstream pipeline take-away limits, and storage-injection rates already close to physical ceilings. The German FSRU programme and the parallel terminal commissioning in Italy, the Netherlands and the Baltics are therefore best read as direct relaxations of the constraint that the SVAR otherwise treats as exogenous. Receiving-side investment does not merely add to total supply security; it changes which margins of adjustment are operationally available at short notice. On the evidence of this exercise, that channel dominates supplier diversification as a marginal-return policy instrument.

The magnitude of Counterfactual 2 should be read as directional rather than literal. The imposed path requires a near-doubling of monthly LNG imports relative to the Feb-22 baseline, well outside the range of historical variation; the linear SVAR overstates demand response at high prices and ignores the upward-sloping global LNG supply curve that would bind at low European prices; the 90% credible band spans more than four log-points, which is itself an honest signal that the model is being asked to extrapolate. The robust content is the *wedge* between the upper and lower envelopes and what that wedge implies about the binding role of receiving-side infrastructure, not either point estimate considered in isolation.

Two methodological notes close the discussion. Across all three counterfactuals the industrial-production gap is small, often of the “wrong” sign relative to the textbook intuition, and statistically indistinguishable from zero; the posterior contemporaneous price-on-output elasticity ($\alpha_{yp} \approx -0.009$) is essentially nil and the sample window is too short for the indirect channels to identify cleanly. Pinning down the real-activity consequences of gas-market disturbances therefore requires complementary evidence beyond this SVAR, ideally from sectoral data or local projections. Counterfactual 3 should likewise be read with care: u^3 bundles policy, the warm-winter anomaly and any residual demand destruction by high prices that exceeds α_{qp} , so the magnitude attaches to the joint signal rather than to any single channel.

5.5.5 Counterfactual 4: a forward-looking scenario fan for the 2026 Hormuz crisis

The first three counterfactuals are retrospective: they ask what would have happened during the 2022–2023 crisis under different supply or demand paths. The fourth counterfactual is forward-looking and addresses a different policy question. The 2026 Strait of Hormuz crisis began on 28 February 2026, with shipping through the Strait substantially blocked and an estimated 20% of global LNG supply removed from the market (International Energy

Agency, 2026; Kiel Institute for the World Economy, 2026). Because our estimation sample ends in March 2026, the first full month of the disruption is part of the data: the realised log-growth of European LNG imports in that month, denoted $\Delta_{\text{Mar-26}}^{\text{LNG}}$, provides a direct empirical anchor for the size of the initial shock. What remains uncertain is the path of the disruption *going forward*: how long the closure binds, whether spillovers to other Gulf chokepoints materialise, and whether the cargo redirection toward Asia becomes a persistent rather than transient feature of the LNG market.

Rather than commit to a single forecast of an inherently uncertain trajectory, we present a fan of three contrasting scenarios, all projected over the twelve months following the end of the sample (April 2026 to March 2027). Each scenario imposes a specific path on g_t^{LNG} over the projection horizon, calibrated against the realised March 2026 shock $\Delta_{\text{Mar-26}}^{\text{LNG}}$:

CF4a – Recovery. LNG net imports return to their pre-Hormuz (February 2026) level over two months and remain stable thereafter. Story: rapid diplomatic or military de-escalation; cargo flows resume within the spring. Imposed path: $g_t^{\text{LNG}} = -\Delta_{\text{Mar-26}}^{\text{LNG}}/2$ in April and May, then zero through March 2027.

CF4b – Persistent. LNG net imports remain at their March 2026 level for the full twelve-month horizon. Story: chronic instability in the Strait, structural redirection of cargoes to Asia, no second wave. Imposed path: $g_t^{\text{LNG}} = 0$ throughout.

CF4c – Worsening. A second wave of comparable magnitude in April, followed by a smaller continuation in May, after which LNG stabilises at the deeper trough. Story: spillover to other Gulf chokepoints, infrastructure damage, regional escalation. Imposed path: $g_t^{\text{LNG}} = \Delta_{\text{Mar-26}}^{\text{LNG}}$ in April, $\Delta_{\text{Mar-26}}^{\text{LNG}}/2$ in May, zero thereafter.

For each posterior draw of the SVAR parameters, the LNG-supply structural shock u_t^{LNG} is set so that the projected g_t^{LNG} matches the scenario path; all other structural shocks are set to zero over the projection horizon, following the deterministic-projection convention of Antolín-Díaz et al. (2021). The 5–95 percent posterior bands therefore reflect parameter uncertainty under the imposed LNG path, not the variability of future non-LNG shocks. The baseline against which the scenarios are compared is the unconstrained SVAR projection from end of sample with all structural shocks set to zero, i.e. the model’s central-tendency forecast. The implied paths for the imposed LNG growth and level series are reported in Figure 13 in the appendix; the full multi-variable response under each scenario, structured as a five-by-three grid (variables by scenarios), is reported in Figure 14.

Figure 7 reports the headline result: the projected path of the real gas price under the three scenarios, overlaid on the observed series. The fan opens cleanly from the end-of-sample anchor. The median price gap (scenario minus baseline projection) at March 2027 is approximately -0.05 log-points under CF4a Recovery, $+0.29$ log-points under CF4b Persistent, and $+0.82$ log-points under CF4c Worsening; the corresponding scenario-versus-baseline level ratios are 0.95, 1.34 and 2.27 respectively. The persistent and worsening scenarios generate posterior bands that exceed the baseline projection by spring 2027 and continue to drift upward as the imposed LNG shortfall compounds through the SVAR’s price equation; the recovery scenario, by contrast, returns to a band overlapping the baseline within a few months.

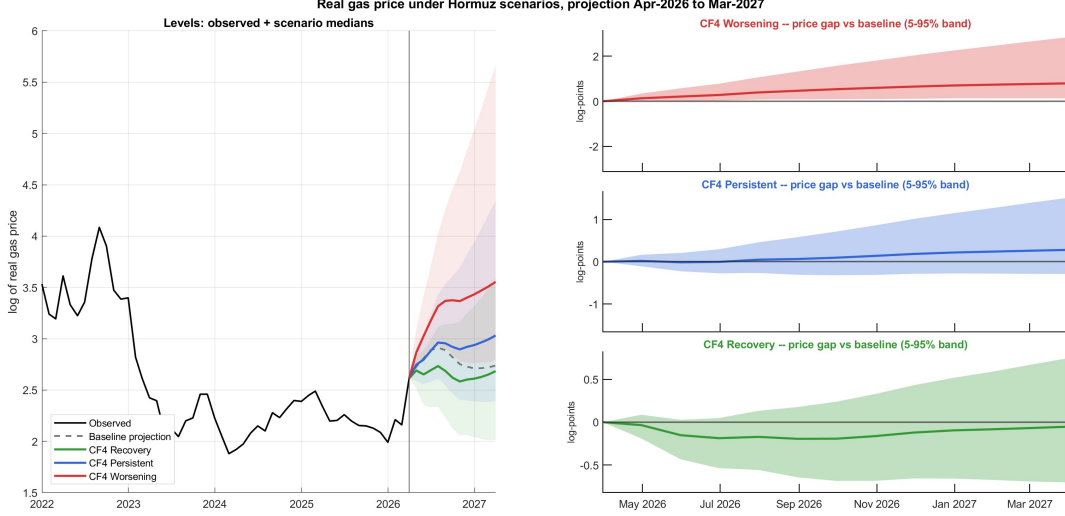


Figure 7: Counterfactual 4: forward-looking fan of three Hormuz scenarios for the real gas price. *Left panel:* levels, observed series (solid black) through March 2026 plus the three scenario medians from April 2026 to March 2027, each with its 5–95% posterior credible band shown in the scenario colour with high transparency; the unconstrained baseline projection is in grey dashed. CF4a Recovery in green, CF4b Persistent in blue, CF4c Worsening in red. *Right panels (stacked):* posterior gap (scenario minus baseline) in log-points for each scenario separately, each with its own 5–95% credible band. The gap bands are anchored at the end-of-sample point (March 2026) where the gap is zero by construction and isolate the scenario-specific price contribution by netting out the common SVAR random-walk component within each posterior draw. The vertical solid line in the left panel marks the end of the estimation sample.

The gap shown in the right panels has a precise interpretation. It is the cumulative log-difference between the scenario and baseline projections at each horizon,

$$\text{gap}_t = \log p_t^{\text{CF}} - \log p_t^{\text{baseline}} = \sum_{s=T+1}^t [g_s^{p,\text{CF}} - g_s^{p,\text{baseline}}],$$

that is, the running sum of the differences between scenario and baseline in monthly log-growth of the real gas price. By construction it is zero at the end-of-sample anchor (March 2026) and accumulates over the projection window; the smooth growth of the gap line in each panel reflects the persistent transmission of LNG-supply shocks to prices that is identified in-sample (rather than a continued sequence of fresh shocks, which would imply discrete jumps in the path). For small values the gap can be read approximately as a percentage difference because $\log(1 + x) \approx x$; for the larger values reached by the Persistent and Worsening scenarios the exact conversion is the level ratio $p_t^{\text{CF}}/p_t^{\text{baseline}} = \exp(\text{gap}_t)$. A gap of +0.29 log-points (Persistent, end-of-horizon) therefore corresponds to a counterfactual price 34% above baseline, and a gap of +0.82 (Worsening) to a price 127% above baseline, or a ratio of $2.27\times$.

Two readings of the fan are worth flagging. First, the price impact of the Hormuz disruption

is sharply asymmetric across scenarios. The recovery scenario converges back to the baseline within a few months, suggesting that the realised March 2026 shock, taken in isolation, is not large enough to substantially shift the medium-run price path if the disruption resolves promptly. The persistent and worsening scenarios, in contrast, generate sustained upward price drift: the inelastic short-run demand identified by the SVAR amplifies an LNG shortfall that does not reverse, with the gap widening over the twelve-month horizon rather than dissipating. The asymmetry mirrors the in-sample finding that LNG-supply shocks have substantially more persistent price effects than pipeline-supply shocks: the same persistence that made LNG the flexible margin during 2022 makes a sustained loss of that margin a persistent price problem.

Second, the policy-relevant magnitude is not the single-point gap at the end of horizon but the conditional distribution across scenarios. From a central-bank perspective, the credible band of CF4b Persistent overlaps materially with that of CF4c Worsening, so the policy-relevant question is not the central forecast but the upper tail of the persistent-or-worse subset. The fan also doubles as a real-time monitoring tool: if observed LNG growth over April–May 2026 exceeds zero, the realised path enters the CF4a Recovery region of the fan and the price implications shrink rapidly; if observed growth tracks at or below zero, the conditional distribution is dominated by CF4b and CF4c, both of which imply sustained price elevation. This conditional-monitoring use of the fan is its principal forward-looking advantage over a single-scenario projection.⁵

6 Conclusions

This paper has studied the European natural gas market through a Bayesian Structural Vector Autoregression that explicitly distinguishes pipeline gas from liquefied natural gas, with contemporaneous elasticities identified through sign and elasticity restrictions. The motivation is empirical: the 2022 energy crisis displayed a level of compositional adjustment in European gas supply that the standard practice of modelling supply as a single aggregate is unable to represent. The contribution of the paper is to take that compositional adjustment seriously, both in the structural model and in the counterfactual exercises that the model makes possible.

The posterior estimates document substantial heterogeneity between the two supply margins. Pipeline-and-domestic supply is highly price-inelastic, with a posterior median elasticity of

⁵Public-domain evidence available at the time of writing (May 2026) places the realised trajectory between the Persistent and Worsening scenarios and substantially closer to the former. QatarEnergy maintained its force-majeure declaration on LNG shipments through April. The first Qatari LNG tanker to transit the Strait since the February closure cleared the northern corridor on 9 May 2026, with Pakistan as its destination rather than Europe; a second tanker followed the next day on the same route. Roughly ten Qatari energy shipments remained held up at that date and the resumption was described as far below pre-war levels of approximately three shipments per day out of the Persian Gulf. The Dutch TTF benchmark remained roughly 35–40% above its pre-closure level through April and May, with a temporary spike of 17–18% in mid-April after US threats of a broader blockade. The Recovery scenario therefore looks empirically discounted; the conditional distribution relevant for the May 2026 outlook is dominated by the Persistent component of the fan.

approximately 0.08, consistent with the contractual rigidities, take-or-pay clauses and infrastructure specificity that characterise this channel. LNG supply is substantially more elastic, with a posterior median of approximately 1.6, consistent with the literature documenting LNG as the flexible balancing margin in Europe. On the demand side, the model identifies a short-run price elasticity that is sizeable in absolute value, an income elasticity in line with cross-country estimates for OECD economies, and a small contemporaneous response of industrial production to gas prices. The structural shocks identified by the model therefore split cleanly into a rigid supply channel, a flexible supply channel, an income-driven demand channel, a price-elastic demand channel, and a storage channel. The historical decomposition shows that the major European price episodes since 2010 arise from the joint action of several of these channels rather than from any single dominant shock.

The impulse-response analysis identifies a clear horizon dependence in gas-price formation. Pipeline supply shocks transmit to prices as sharp but transient spikes that largely dissipate within five months, whereas LNG supply shocks transmit more persistently by shifting the medium-run market-clearing condition. The forecast-error variance decomposition assigns roughly comparable short-horizon shares to supply and demand shocks, and identifies storage-related precautionary demand as a significant amplification channel during episodes of supply stress.

The three counterfactual exercises bracket the contribution of the two main adjustment mechanisms identified by the model during the 2022–2023 crisis. Counterfactuals 1 and 2 operate on the LNG supply margin: holding LNG imports frozen at their February 2022 level over the crisis window produces a median price gap of approximately +0.80 log-points (Counterfactual 1), so that prices would have been roughly 2.2 times the observed level; allowing LNG to fully absorb the contemporaneous pipeline-and-domestic shortfall produces a gap of -1.70 log-points (Counterfactual 2), implying a counterfactual price of roughly 18% of the observed level, only modestly above the pre-crisis normal range. Counterfactual 3 operates on the demand side: eliminating the autonomous demand-reduction signal identified by the SVAR over the formal EU policy window of Council Regulation 2022/1369 produces a gap of approximately +0.30 log-points, implying that prices would have been about 35% higher, persistently, had the joint contribution of the policy package, the warm winter 2022–23 and the price-driven demand destruction not materialised. The realised price path lies closer to the LNG-frozen upper envelope than to the LNG-unconstrained lower envelope, an asymmetry consistent with the regasification bottleneck having bound during the crisis. The wedge between observed prices and the Counterfactual 2 trajectory is naturally interpreted as the implicit cost of that bottleneck: globally, LNG cargoes were redirectable, but Europe’s capacity to receive, regasify and distribute additional volumes was operationally constrained. The German FSRU programme and the parallel expansions of regasification capacity elsewhere in Europe are most naturally read as direct relaxations of the constraint that this exercise quantifies.

The magnitude reported for Counterfactual 2 should be read as a qualitative lower bound rather than a literal forecast. The SVAR is linear and is estimated on a sample in which monthly LNG growth varied within a range an order of magnitude smaller than the shock the counterfactual imposes. Three sources of model misspecification push the literal point

estimate away from any plausible realised outcome of an unconstrained-LNG scenario: gas demand becomes more elastic at high prices and less elastic at low prices, so the linear demand curve overstates the response at the tails; LNG cargo supply to Europe is itself bounded, since at sufficiently low European prices cargoes redirect to Asia; and posterior uncertainty about the structural elasticities translates into a wide credible interval whose width itself signals that the model is being asked to extrapolate. The robust content of the exercise is the wedge between the three scenarios and what it implies about the binding role of receiving-side infrastructure, rather than any one point estimate considered in isolation.

The forward-looking exercise in Section 5.5.5 applies the same model to the 2026 Strait of Hormuz disruption. Because the estimation sample now ends in March 2026, the first full month of the disruption is part of the data and provides an empirical anchor for the size of the initial LNG shock; what is left to model is the path of the disruption going forward. The fan of three contrasting scenarios over a twelve-month horizon delivers a sharply asymmetric price response: prompt resolution returns prices to within 5% of the baseline projection by March 2027, whereas persistent disruption produces a sustained drift of $\approx +34\%$ and an escalating second-wave scenario reaches $\approx +127\%$ (a $2.3\times$ ratio). The asymmetry is driven by the inelastic short-run demand identified by the SVAR, which amplifies an LNG shortfall that does not reverse. The fan doubles as a real-time monitoring instrument, with observed LNG growth over April–May 2026 placing the realised path within one of the three scenario regions; the public-domain evidence available at the time of writing locates that realised trajectory between the persistent and worsening scenarios, considerably closer to the former. The exercise also points to a complementary policy lever to the receiving-side infrastructure emphasised by the retrospective counterfactuals: geographic diversification of LNG supply across producing regions and shipping chokepoints, of which the concentrated Qatar–Hormuz dependence is the most visible example, is a first-order determinant of how resilient European prices are to localised disruptions abroad.

The broader implications cut across energy policy, central-bank practice and applied modelling. Energy security in Europe is not only a function of supply level or supplier diversity but of the receiving-side infrastructure that decides which adjustment margins are operationally available at short notice; the marginal-return policy instrument identified by these counterfactuals is investment in regasification, downstream pipeline take-away and storage-injection capacity, not in supplier diversification alone. For monetary policy, the horizon-dependent structure of gas-price responses is directly actionable: a pipeline-led spike dissipates within roughly five months on the evidence reported here and can credibly be treated as a transient headline-inflation deviation, whereas an LNG-led shock shifts the medium-run market-clearing condition and is more likely to feed into core inflation through energy-intensive intermediate goods, into wage- and price-setting through expectations, and into second-round effects through sectoral pass-through. The composition of a gas-market disruption is therefore a first-order input to the calibration of the policy response, and the framework developed here provides a structural way to extract that composition from the data. The inference exercise itself argues for taking supply composition seriously: aggregating pipeline gas and LNG into a single supply variable obscures exactly the substitution dynamics that drive the most consequential price episodes.

Several avenues for further work suggest themselves. Extending the framework to explicitly model international spillovers through global LNG markets, incorporating forward prices and futures-based measures of expectations, or embedding the identified shocks into a broader macro-financial model would enrich the understanding of how energy-driven fluctuations transmit to inflation, output and equity returns. Higher-frequency data or sectoral demand disaggregation may help sharpen inference on the adjustment mechanisms during periods of acute stress; complementary evidence from local projections may help isolate the real-activity consequences of gas-market shocks, which the SVAR analysed here is unable to identify with confidence within its sample. Beyond these methodological extensions, the most immediate empirical extension is the retrospective analysis of the 2026 disruption: once JODI coverage is complete, the contemporaneous elasticities and the counterfactual framework developed here can be applied directly to the new episode, providing both a cross-validation of the present results and a quantification of a second European gas crisis through a unified empirical lens.

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A Data treatment at the bleeding edge of the sample

The estimation sample ends in March 2026, which is also the first full month of the Hormuz crisis (started 28 February 2026). The last few observations rely on preliminary releases that are subject to subsequent revision in three ways. First, JODI Gas reports four small countries (EE, IE, LU, SI) with assessment code 3 (*Imputed*) for the monthly stock change (**STOCKCH**) in March 2026; rather than use JODI’s own imputation, we reconstruct their stock change from the country-level first-difference of the closing stock level (**CLOSTLV**),

$$\widehat{\text{STOCKCH}}_{c,t} = \text{CLOSTLV}_{c,t} - \text{CLOSTLV}_{c,t-1},$$

an accounting identity that uses only data reported with code ≤ 2 by the national authorities of those four countries. They jointly represent less than 3% of EU-27 gas consumption, so the quantitative effect on the aggregate **STOCKCH** series is negligible. Second, four larger countries (BE, DE, FI, NL) submit some required flows (notably **INDPROD**, **EXPPIP** and **EXPLNG**) with code 2 (*Preliminary*) in March 2026; we include these because JODI’s standard practice reclassifies preliminary entries to Final in subsequent monthly releases, and historical revisions between Preliminary and Final vintages have been small in our 2010–2025 sample. Third, the Eurostat HICP and the World Bank Pink Sheet entries for March 2026 are also preliminary releases and may revise marginally in the next vintage. The empirical results presented in the main text are robust to dropping the last observation; we report the full sample because the Hormuz episode is precisely the event we want the estimated dynamics and counterfactual exercises to incorporate.

B Imposed LNG path under Counterfactual 1

Figure 8 reports the implied path of net LNG imports under the counterfactual versus the observed series. The left panel shows the level (JODI units): from February 2022 onward the counterfactual remains flat at the freeze value, while the observed series continues to grow toward its 2024 peak. The right panel shows the same series in monthly log-growth: the counterfactual is identically zero throughout the freeze window. After December 2023 the counterfactual returns to the observed values; the model is then run forward without further constraints and the structural innovations to the LNG-supply equation are once again historical.

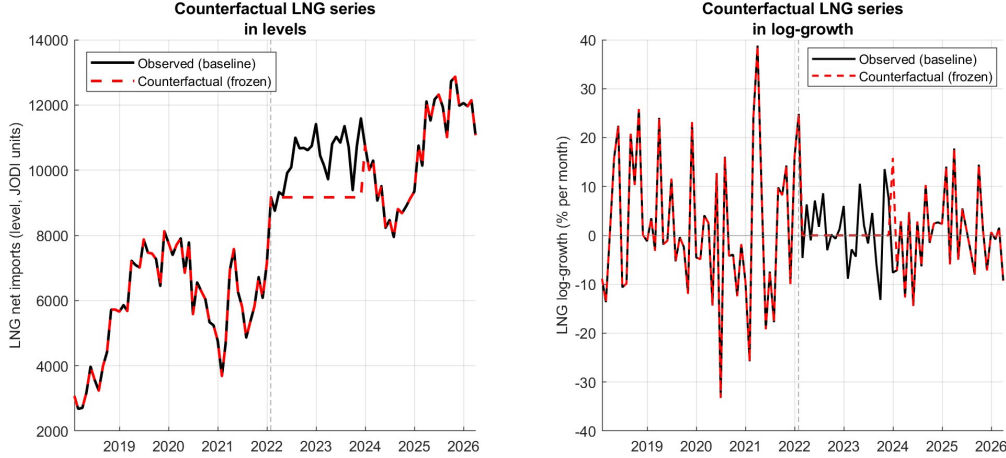


Figure 8: Counterfactual 1: imposed path of net LNG imports. *Left*: level, observed (solid black) versus counterfactual (dashed red). *Right*: monthly log-growth of net LNG imports (in percent per month), observed (black) versus counterfactual (red). The vertical dashed line marks February 2022, the start of the freeze window.

C The structural demand shock u^3 under Counterfactual 3

Figure 9 reports the time series of the third structural shock u^3 as identified by the SVAR, together with its counterfactual counterpart in which the shock is set to zero during the policy window. The left panel shows the posterior median of u^3 across all $\sim 500,000$ retained draws, with the 5–95% credible band shaded in grey; the right panel shows the cumulative sum from the start of the sample, which makes the persistence of the autonomous demand reduction visually salient. The vertical dashed lines mark the start (1 August 2022) and end (31 March 2023) of the formal EU policy window under Council Regulation 2022/1369. During this window the posterior median of u^3 takes systematically negative values, indicating that the SVAR’s demand equation absorbs a non-trivial unexplained downward shift in gas demand, the empirical signature of the joint policy, temperature and behavioural-conservation response discussed in Section 5.5.3. The counterfactual replaces this realised sequence with zeros over the same months and leaves the rest of the series unchanged.

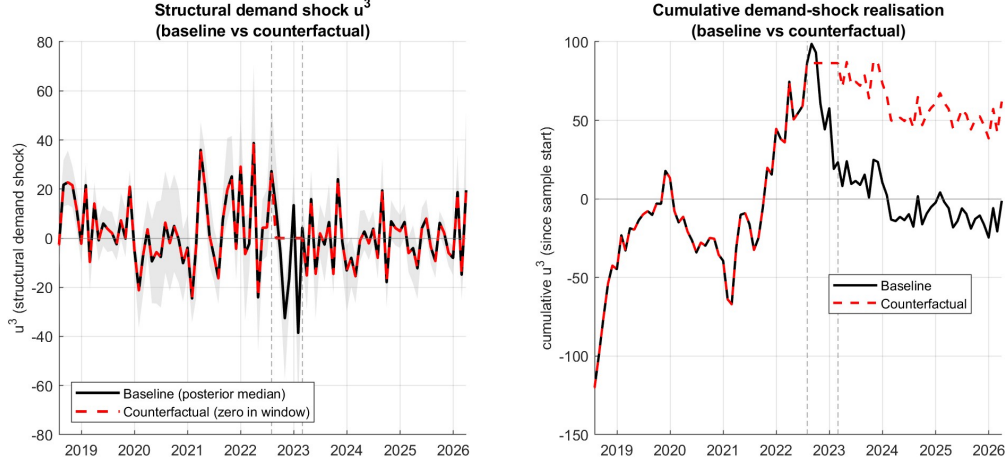


Figure 9: Counterfactual 3: identified structural demand shock u^3 . *Left*: posterior median of u^3 over the sample (solid black) and its counterfactual replacement (dashed red, zero in the policy window); the shaded band is the 5–95% posterior credible interval of the baseline series. *Right*: cumulative sum of u^3 since the start of the sample, baseline (black) versus counterfactual (red), which illustrates the persistence of the autonomous demand-reduction signal. The vertical dashed lines mark 1 August 2022 and 31 March 2023, the start and end of the formal policy window under Council Regulation 2022/1369.

D Imposed LNG path under Counterfactual 2

Figure 10 reports the implied path of net LNG imports under Counterfactual 2 versus the observed series. The counterfactual is anchored at the February 2022 LNG volume and rises one-for-one with the contemporaneous pipeline-and-domestic shortfall, $\text{LNG}^{cf}(t) = \text{LNG}(\text{Feb-22}) + [q^{pip}(\text{Feb-22}) - q_{\text{obs}}^{pip}(t)]$. The left panel shows the level (JODI units): from February 2022 onward the counterfactual climbs sharply as it absorbs the pipeline shortfall, peaking at roughly twice its Feb-22 level (and substantially above the observed series) during 2023 before returning to the observed series in 2024. The right panel shows the same series in monthly log-growth: the counterfactual deviates from the observed path only during the absorption window, with positive deviations in 2022–2023 as LNG ramps up to offset the pipeline gap, and a sharp negative correction at the end of the window when the constraint is released and the counterfactual rejoins the observed series.⁶ The implied counterfactual LNG volumes during 2022–2023 substantially exceed the regasification, downstream pipeline and global cargo availability that were operationally feasible at the time, which is precisely the physical bottleneck that the scenario abstracts away from.

⁶The sharp drop in the log-growth panel at the end of the window is a mechanical artefact of forcing the counterfactual level back to the observed series in a single month; the underlying *level* path is smooth.

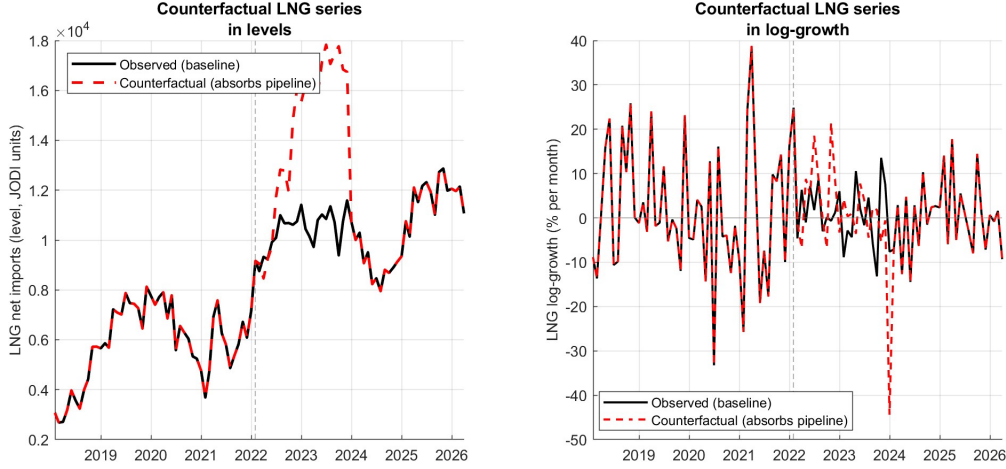


Figure 10: Counterfactual 2: imposed path of net LNG imports. *Left*: level, observed (solid black) versus counterfactual (dashed red, “absorbs pipeline”). *Right*: monthly log-growth of net LNG imports (in percent per month), observed versus counterfactual. The vertical dashed line marks February 2022, the start of the absorption window.

E Counterfactual 1 with LNG growing at its pre-crisis trend

The baseline of Counterfactual 1 freezes net LNG imports at their February 2022 volume, a deliberately extreme upper bound. A milder variant constrains LNG growth not to zero but to its pre-crisis trend $\bar{g}_{\text{pre-crisis}}^{\text{LNG}}$, computed as the mean monthly log-growth of net LNG imports over January 2017–December 2021 (≈ 1.19 pp per month, equivalent to roughly 14% annualised). The constraint $g_t^{\text{LNG}} = \bar{g}_{\text{pre-crisis}}^{\text{LNG}}$ is imposed over February 2022–December 2023 via the same forward-simulation procedure used for Counterfactual 1. The exercise isolates the marginal contribution of the crisis-induced “policy push” (FSRUs, EU–US task force, accelerated regasification commissioning) from the secular growth trajectory already in place.

The median price gap rises to approximately +0.20 log-points during the constraint window and then drifts negative in the post-2023 period, settling at −0.20 log-points by the end of the sample (90% credible interval roughly [−0.45, +0.05]). The IP and inventory gaps remain small throughout. The mechanics behind the sign are informative. Imposing the pre-crisis trend of ~ 1.19 pp per month over 22 months delivers a cumulative LNG growth of ~ 26 pp; the realised cumulative LNG growth over the same window was ~ 23 pp. The trend counterfactual therefore puts *more* LNG into the system than was actually achieved, and the resulting post-constraint dynamics pull prices below baseline through the standard supply-elasticity channel. Two readings follow. First, the realised LNG expansion during 2022–2023 was below, not above, what continuing the pre-crisis trend would have delivered cumulatively; the crisis-induced policy push was therefore necessary at the margin, not redundant. Second, the magnitudes are small either way: the baseline Counterfactual 1 (LNG fully frozen) produces +0.80 log-points, an order of magnitude larger than what is at

stake when one merely interpolates between trend continuation and observed expansion. The stabilising role of the LNG margin during the crisis came primarily from the accumulated stock of receiving infrastructure built up over the preceding decade, with the rapid-response policy interventions of 2022 contributing a modest but not negligible additional amount.

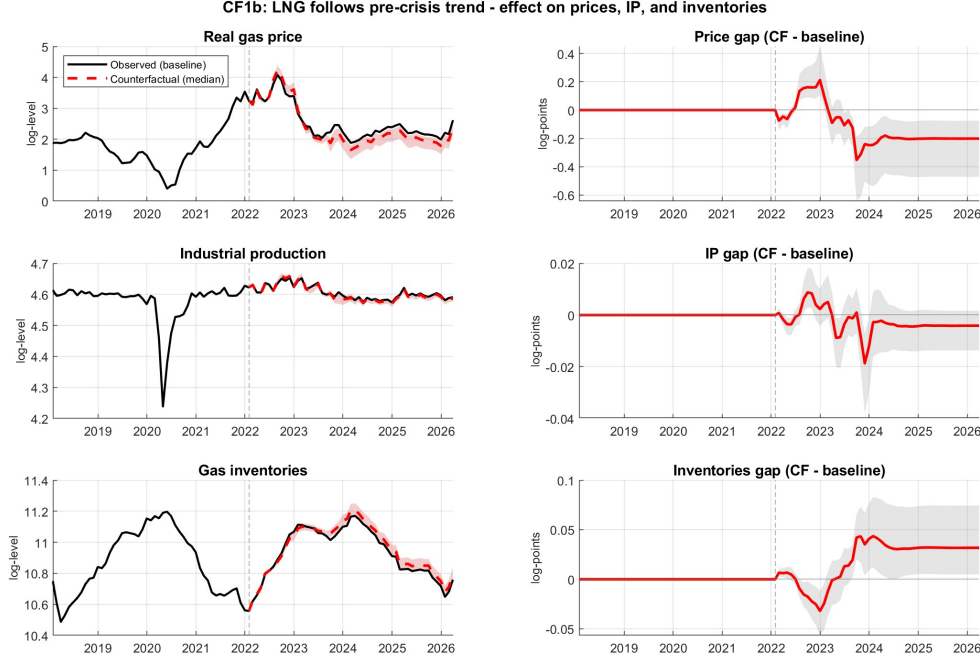


Figure 11: Counterfactual 1b: full multi-variable response under $g_t^{LNG} = \bar{g}_{\text{pre-crisis}}^{LNG}$. Each row: one endogenous variable; left: log-level (observed solid black, counterfactual median dashed red, 5–95% band shaded); right: posterior gap with the same band, in log-points. Vertical dashed line marks February 2022.

F Counterfactual 3 over an extended window

The baseline Counterfactual 3 sets $u_t^3 = 0$ over the formal Council Regulation 2022/1369 window (August 2022–March 2023). We re-run the exercise with the constraint window extended to December 2023, motivated by (i) the formal extension of the regulation through Council Regulation 2023/706 (valid until 31 March 2024) and (ii) the cumulative path of u^3 in Figure 9, which continues to drift downward through the end of 2023.

The extended-window result is broadly comparable to the baseline. The median price gap stabilises at approximately +0.30 log-points by the end of the sample, against +0.30 in the baseline specification, with a 90% credible interval of similar width. The demand-reduction signal identified by the SVAR therefore carries through the prolonged constraint window without significant offsetting innovations: the cumulative contribution of the autonomous demand-reduction component is similar whether one stops at March 2023 or continues through December 2023. The robustness of the magnitude to this window choice strengthens the baseline interpretation: it is the cumulative demand-reduction signal that drives the price contribution, not a specific calendar boundary that one happens to choose.

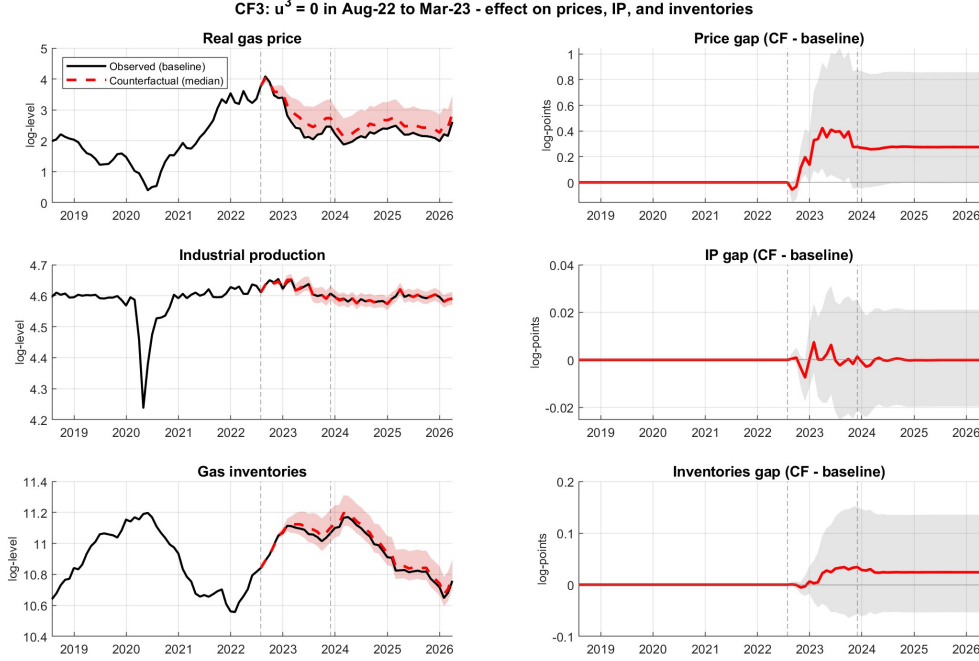


Figure 12: Counterfactual 3 over an extended window: full multi-variable response under $u_t^3 = 0$ over August 2022–December 2023. Same format as Figure 11. The two vertical dashed lines mark the start and end of the extended constraint window.

G Counterfactual 4: imposed LNG paths under the three scenarios

Figure 13 reports the imposed LNG paths that define the three scenarios of Counterfactual 4. The left panel shows the imposed monthly log-growth (pp/month) of LNG over the twelve-month projection horizon (April 2026 to March 2027); the right panel shows the implied LNG level (posterior median across MCMC draws) overlaid on the observed series. The three scenarios are calibrated symmetrically around the realised March 2026 LNG log-growth $\Delta_{\text{Mar-26}}^{\text{LNG}}$: Recovery (green) imposes two months of positive growth that exactly offset the March drop and then zero growth thereafter; Persistent (blue) imposes zero growth throughout, holding LNG at its March 2026 level; Worsening (red) repeats the March-magnitude shock in April, a half-magnitude continuation in May, and then zero growth at the deeper trough. By construction, all three scenarios depart from the same end-of-sample anchor and differ only in their subsequent paths; the LNG-level panel illustrates how this translates into the cumulative quantitative position of the LNG margin under each scenario.

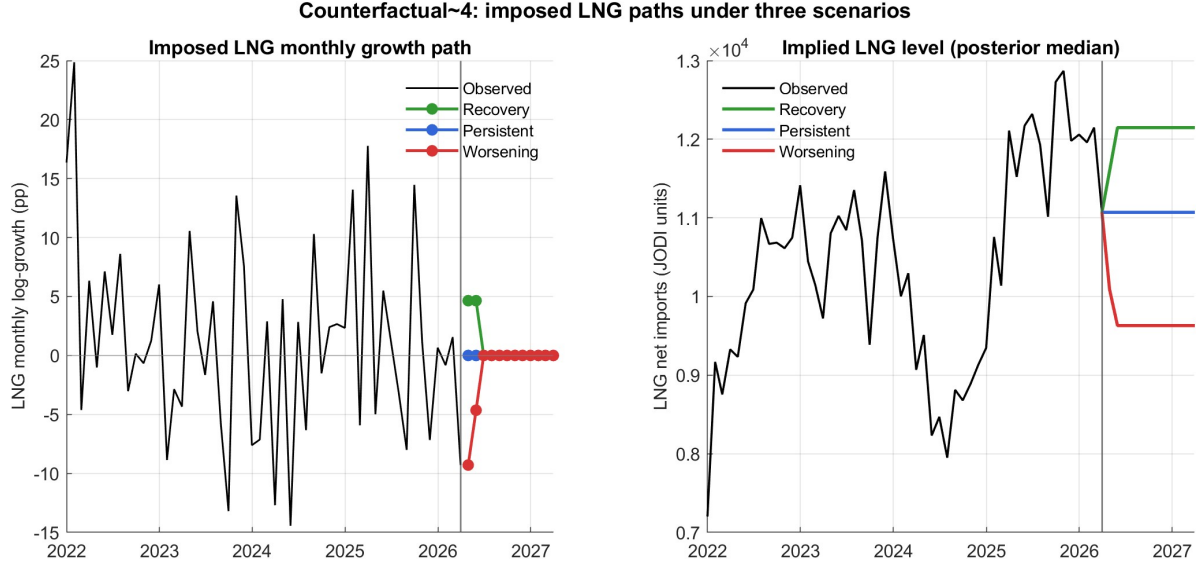


Figure 13: Counterfactual 4: imposed LNG paths under the three scenarios. *Left*: imposed monthly log-growth of LNG (pp/month) over April 2026–March 2027; observed series in black, CF4a Recovery (green), CF4b Persistent (blue), CF4c Worsening (red). *Right*: implied LNG level (posterior median across draws) overlaid on the observed series. The vertical solid line marks the end of the estimation sample (March 2026).

H Counterfactual 4: full multi-variable response under the fan

Figure 14 reports the full multi-variable response of the SVAR under each of the three Counterfactual 4 scenarios, structured as a five-by-three grid: rows index the five endogenous variables of the system (q^{pip} , industrial production, real gas price, inventories, q^{LNG}), and columns index the three scenarios (CF4a Recovery, CF4b Persistent, CF4c Worsening). Each panel reports the observed in-sample path (solid black), the unconstrained baseline projection (grey dashed), and the scenario median (solid colour) with the 5–95% posterior credible band. The figure thus disaggregates the headline price fan of Figure 7 into its component responses across the rest of the system: how the price elevation under CF4b and CF4c is accompanied by inventory drawdowns and modest reductions in industrial production; how the pipeline supply margin responds endogenously to the LNG shortfall through the estimated cross-equation feedback; and how the recovery scenario brings all variables back toward the baseline projection within a few months. The figure is intended as a reference for readers interested in the full posterior implications of the fan, beyond the headline price result reported in the main text.

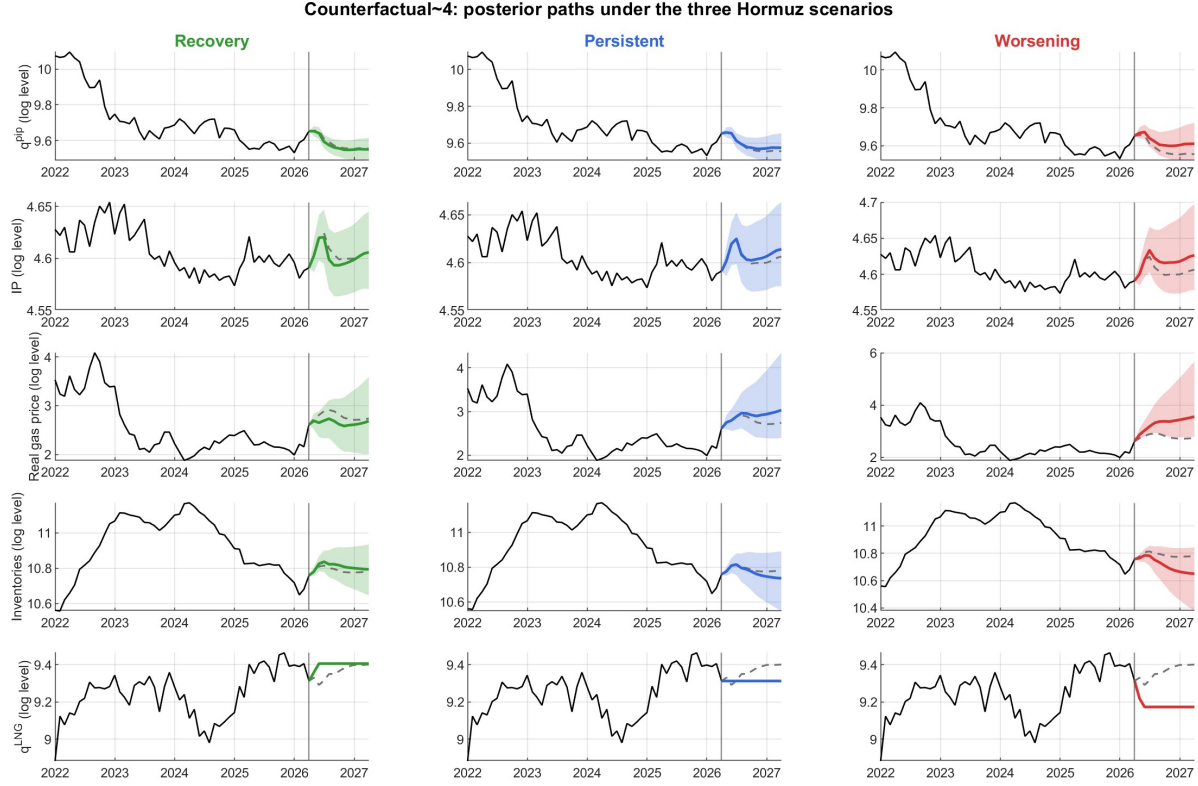


Figure 14: Counterfactual 4: full multi-variable response under the three Hormuz scenarios. Rows: the five endogenous variables (q^{pip} , industrial production, real gas price, inventories, q^{LNG}); columns: the three scenarios (CF4a Recovery, CF4b Persistent, CF4c Worsening). Each panel reports the observed series (solid black), the unconstrained baseline projection (grey dashed), and the scenario median (solid colour) with the 5–95% posterior credible band. The vertical solid line in each panel marks the end of the estimation sample (March 2026).

I Robustness: IEA Gas Trade Flows in place of JODI

This appendix re-estimates the structural SVAR on an alternative source for the two supply quantities. The identifying restrictions and prior specification are unchanged; the only difference is that pipeline-and-domestic supply (q^{pip}) is replaced by the IEA Gas Trade Flows pipeline aggregate (Russia + Norway + Algeria + Iran + Turkey, all routes summed), and LNG (q^{LNG}) is replaced by the IEA LNG aggregate. Both IEA series are seasonally adjusted with X-13 ARIMA-SEATS, identically to the JODI baseline. The sample window is the same.

The two data sources measure related but not identical objects. JODI sums national reports of customs-recorded flows for the 22 EU member states with complete coverage. IEA Gas Trade Flows tracks commercial cargo and pipeline-route arrivals at the European level. Differences in reporting frequency, geographic coverage and the exact timing convention (when a cargo is counted as arrived versus when it enters the national distribution network)

translate into different month-to-month dynamics for the two series. Figure 15 reports the two pipeline series and the two LNG series side by side, in cumulative log-level and in monthly log-growth, over the SVAR sample.

The pipeline cumulative log-levels in Figure 15 (top-left panel) drift apart from 2012 onward. The chart anchors both series at zero at the start of the sample (August 2010) and accumulates monthly log-growth from there, so what the panel shows is not absolute volume but cumulative growth relative to that anchor: in absolute terms (bcm), JODI is larger than IEA in any given month because it includes European domestic production on top of pipeline imports, but in cumulative log-growth IEA sits above JODI because JODI's growth rate is dragged down by the secular decline of domestic production. The Dutch Groningen field, the dominant domestic producer over the sample, fell from roughly 50 bcm per year in 2013 to a near-zero post-earthquake phase-out by 2024; that secular decline shows up in JODI's pipeline-and-domestic aggregate but not in the IEA pipeline aggregate, which counts external producer-country flows (Russia, Norway, Algeria, Iran, Turkey) only. The post-2022 break is visible in both series as a sharp drop in growth, but JODI declines more sharply because the IEA aggregate does not include the Groningen-driven secular trend at all. Differences in geographic scope (UK and Turkey treatment, LNG re-exports between EU countries) and in intra-EU netting also contribute, but the domestic-production differential is the largest single factor visible in the top-left panel of the figure.

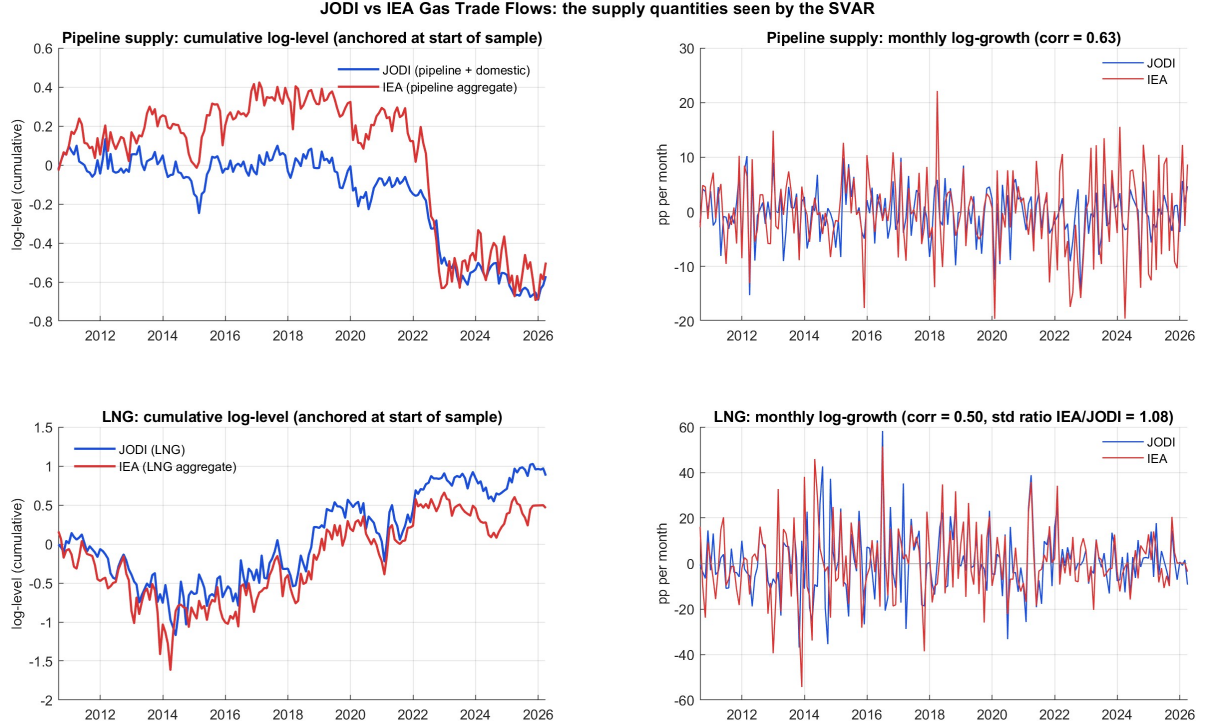


Figure 15: JODI baseline (blue) versus IEA Gas Trade Flows (red) for the two supply aggregates as they enter the SVAR. *Top row*: pipeline supply, in cumulative log-level and in monthly log-growth. *Bottom row*: LNG, in cumulative log-level and in monthly log-growth. The two pipeline aggregates differ in coverage (IEA is the sum of commercial pipeline routes only; JODI is net pipeline imports plus domestic production for the 22 EU member states) and the IEA series is correspondingly more volatile in monthly log-growth (std 7.2 pp vs 4.7 pp, correlation 0.63). The two LNG aggregates have similar overall volatility (std 15.4 pp vs 14.2 pp) but the month-to-month correlation between them is only 0.50, indicating they capture different components of LNG-flow variation rather than a smoothed version of the same series.

Table 5 reports the posterior median and the 5–95% credible interval for each of the six key contemporaneous elasticities under the two data sources. Three observations follow. First, every sign is preserved: pipeline-and-domestic supply remains the rigid margin and LNG the flexible one, demand responds positively to income and negatively to its own price (mapped to a positive α_{qp} in our convention), inventories respond negatively to price, and the contemporaneous response of industrial production to price is small. Second, the IEA estimates compress the ranking somewhat: the LNG-supply elasticity falls from $\beta_{qp} \approx 1.4$ to ≈ 0.6 , while the pipeline-supply elasticity rises from $\gamma_{qp} \approx 0.08$ to ≈ 0.18 . The flexible-rigid ratio shrinks from roughly $17\times$ to $3.4\times$, but the qualitative conclusion—LNG materially more elastic than pipeline—remains in place. Third, the demand-side elasticity α_{qp} is roughly three times larger in the IEA variant; the auxiliary elasticities (α_{qy} , ψ_2 , α_{yp}) are very similar across specifications.

Table 5: Posterior medians and 5–95% credible intervals for the contemporaneous elasticities of the SVAR, JODI baseline against IEA robustness. Both runs use the same identifying restrictions and the same prior specification.

Parameter	JODI baseline	IEA robustness
β_{qp} (LNG supply on price)	+1.365 [+0.64, +2.80]	+0.608 [+0.25, +1.17]
γ_{qp} (pipe supply on price)	+0.081 [+0.04, +0.12]	+0.176 [+0.11, +0.25]
α_{qp} (demand on price)	+0.590 [+0.30, +1.27]	+1.702 [+0.96, +3.28]
α_{qy} (demand on IP)	+0.874 [+0.43, +1.55]	+0.849 [+0.37, +1.65]
ψ_2 (inventory on price)	−0.057 [−0.09, −0.02]	−0.047 [−0.09, −0.01]
α_{yp} (IP on price)	−0.011 [−0.03, −0.00]	−0.010 [−0.03, −0.00]

The most economically interpretable difference is the LNG-supply elasticity. The two LNG series in Figure 15 have similar overall volatility but only 0.50 correlation in monthly log-growth: they measure the same economic concept but capture different timings of the variation. The JODI series records LNG made available to the national distribution networks and so incorporates regasification scheduling, storage-injection timing and downstream redelivery, all of which can re-shuffle between months in response to contemporaneous price signals. The IEA series counts cargoes physically arriving at terminals; those cargoes are contracted, scheduled and dispatched weeks to months in advance, so the contemporaneous price has only a small effect on what physically arrives within the same calendar month. The two estimates of β_{qp} therefore identify the same underlying supply margin at two different points along its adjustment chain: JODI captures the regasification-and-redelivery component that responds within the month, whereas IEA captures only the residual contemporaneous response of vessel arrivals. Reading the two estimates together, the true monthly elasticity of European LNG supply to price most likely sits between the two values reported here, closer to 1.0.

The impulse-response comparison in Figure 16 confirms the same picture in dynamic form. Across the 5×5 grid of cumulative responses, the JODI and IEA posteriors share the same qualitative shape: pipeline-supply shocks transmit as transient price spikes that dissipate within a handful of months; LNG-supply shocks transmit as more persistent shifts in the price level; demand and storage shocks display the patterns documented for the baseline. Where the median responses differ, the differences sit comfortably inside the 90% credible bands. The economically substantive conclusions of the paper—horizon-dependent transmission, supply-margin heterogeneity, the policy implications for receiving-side infrastructure and supplier diversification—are robust to the choice of data source.

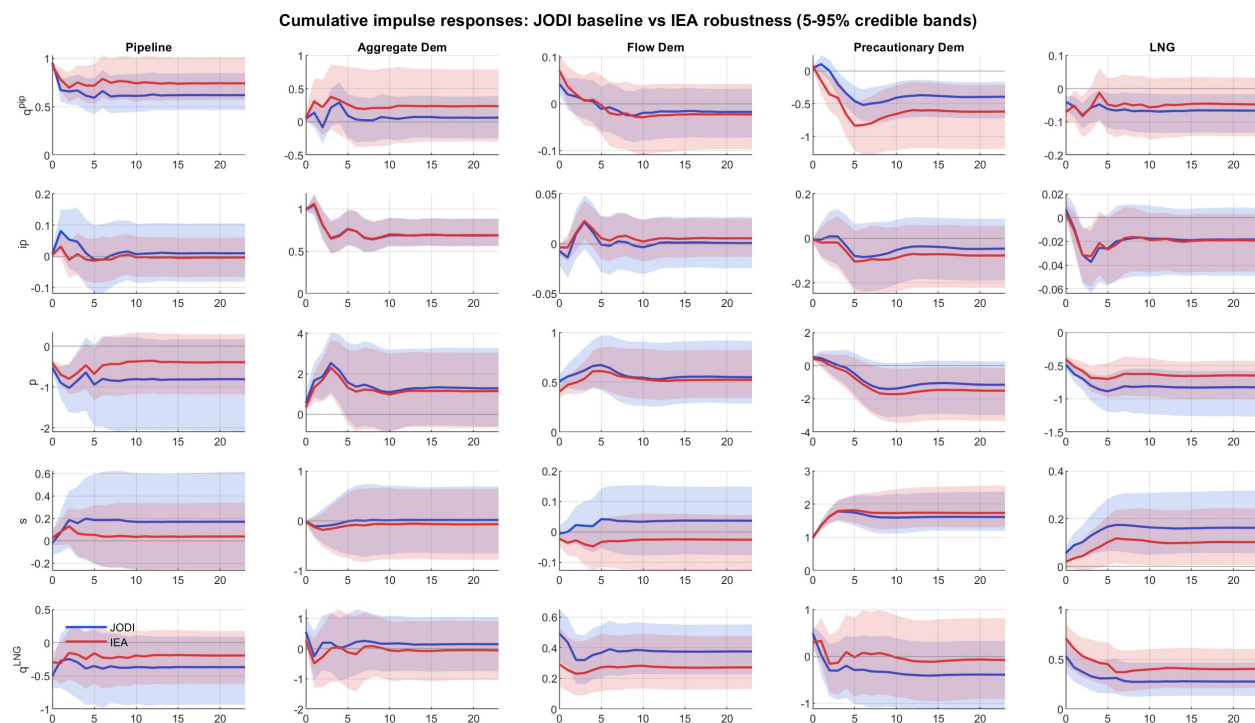


Figure 16: Cumulative impulse responses of the five endogenous variables (rows) to the five structural shocks (columns), JODI baseline (blue) and IEA robustness (red). Solid lines are the posterior medians; shaded areas are the 5–95% credible bands. The horizon is 24 months.